



Monetary Authority of Singapore

Guidelines
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Guidelines on Liquidity Risk Management (Banks)



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Guidelines on Liquidity Risk Management (Banks)

1 Introduction

- 1.1 These Guidelines set out MAS' expectations on liquidity risk management practices for banks, merchant banks and finance companies (hereafter collectively referred to as "Banks") supervised by the Monetary Authority of Singapore (MAS). They are based on the Principles for Sound Liquidity Risk Management and Supervision issued by the Basel Committee on Banking Supervision (BCBS)¹ in 2008 and incorporate insights gained from the banking events in March 2023².
- 1.2 These Guidelines complement existing legislation and guidelines issued by MAS³. They should be read in conjunction with relevant guidelines, information papers and circulars that MAS may issue from time to time. Relevant industry standards should also be considered where appropriate. The Guidelines apply on a group basis for locally-incorporated Banks⁴.
- 1.3 A Bank should adopt the Guidelines in a manner commensurate with the nature, size and complexity of its activities. The good practices in boxed items illustrate practices by certain Banks on how they meet the expectations in the Guidelines. They serve as references for a Bank to consider when determining appropriate controls for its specific situations.
- 1.4 MAS recognises that liquidity risk may be managed on a group-wide basis. A foreign bank branch may leverage its head office's group-wide liquidity risk management framework, provided that it demonstrates that the framework, policies and processes meet the expectations set out in the Guidelines and adequately address Singapore-specific considerations. In all cases, local senior management is expected to have adequate visibility into and input on matters affecting

¹ <https://www.bis.org/publ/bcbs144.pdf>

² The period witnessed deposit runs at multiple banks, where outflow rates were more severe than past stressful events.

³ The examples provided here are non-exhaustive:

- For banks and merchant banks, MAS Notice 649 and MAS Notice 1015 on Minimum Liquid Assets and Liquidity Coverage Ratio, MAS Notice 652 on Net Stable Funding Ratio.
- For finance companies, MAS Notice 806 on Minimum Cash Balances and Minimum Liquid Assets.

⁴ For a locally-incorporated Bank that is headquartered in Singapore, this refers to the group including the holding company in Singapore, as well as the Bank's subsidiaries and branches, in Singapore and overseas, where applicable. For a locally-incorporated subsidiary of a foreign bank, this refers to the subsidiary's operations in Singapore and its downstream subsidiaries and branches, in Singapore and overseas, where applicable.

Singapore operations, supported by timely and relevant reports and data. Clear escalation paths should exist for Singapore-specific issues to be surfaced to the responsible parties to ensure timely decision-making and resolution. Local senior management should also be able to demonstrate to MAS how it discharges its oversight responsibilities, even when relying on group frameworks.

2 Fundamentals

- 2.1 There are two types of liquidity risk. Funding liquidity risk is the risk that a Bank will not be able to efficiently meet both expected and unexpected current and future cash flows and collateral needs without affecting either daily operations or the financial condition of the Bank. Market liquidity risk is the risk that a Bank cannot easily offset or eliminate a position at the market price because of inadequate market depth or market disruption. These Guidelines focus on funding liquidity risk. However, it is recognised that market liquidity risk can significantly affect funding liquidity risk and should be considered where relevant.
- 2.2 Mismanagement of liquidity risk can potentially lead to a Bank failing to meet its financial obligations, thereby undermining public and market confidence in the Bank. A crisis of confidence arising from poor management of business activities or other forms of risks can also trigger liquidity stress events. Robust risk management practices reduce the likelihood of liquidity shortfalls and buy valuable time for a Bank to address any underlying issues. A Bank should therefore establish well-defined frameworks and processes to maintain a sound funding profile and sufficient liquidity to meet obligations as they fall due, and to build resilience against stress scenarios, including severe ones.

3 Roles and Responsibilities of Board and Senior Management

- 3.1 A Bank is responsible for the sound management of its liquidity risk. Strong oversight by the board of directors⁵ and senior management is critical to effective liquidity risk management.

⁵ For a Bank incorporated in Singapore, this refers to the board of directors or a board-level committee which is delegated by the board of directors. For a Bank that is incorporated outside Singapore, this could be a governing body or committee beyond

3.2 The board is ultimately responsible for the liquidity risk assumed by the Bank and the manner in which this risk is managed. It is responsible for:

- (a) approving the Bank's liquidity risk appetite;
- (b) approving and regularly reviewing (at least annually) the Bank's liquidity strategies, key policies, procedures and practices (e.g., liquidity stress test and contingency funding plan);
- (c) overseeing senior management's implementation of these liquidity strategies, policies, procedures and practices to ensure effective liquidity risk management that aligns with the established liquidity risk appetite;
- (d) staying informed about the Bank's overall liquidity profile by reviewing regular reports on its liquidity position (e.g., stress test results) and requiring senior management to promptly update on new or emerging liquidity concerns;
- (e) ensuring that an appropriate organisational structure is established for managing liquidity risk;
- (f) ensuring that senior management and appropriate personnel possess the necessary expertise to manage liquidity risk effectively; and
- (g) ensuring that senior management takes appropriate and prompt remedial actions to address any concerns noted.

3.3 Senior management is responsible for:

- (a) developing and implementing liquidity strategy, policies, procedures and practices aligned with board-approved risk appetite and ensuring sufficient liquidity. This includes implementation of a sound process for identifying, measuring, evaluating, monitoring, reporting and controlling or mitigating liquidity risk;
- (b) establishing the structure⁶, responsibilities and controls for overseeing and managing liquidity risk and liquidity positions of all legal entities, branches and subsidiaries, and outlining these elements clearly in the Bank's liquidity policies;
- (c) communicating liquidity strategy, policies, procedures and management structure to relevant parties for implementation;
- (d) establishing reporting requirements, including scope, manner, and frequency of reporting for various recipients (such as the board, senior management and relevant committees), as well as assigning responsibilities to parties to prepare the reports;
- (e) defining procedures and approving authorities for policy exceptions, including escalation and

local management that is responsible for the oversight of the Bank in Singapore, such as a regional or global committee which exercises oversight of the liquidity and funding risks of the Singapore branch.

⁶ For instance, the degree of centralisation or decentralisation of a Bank's liquidity risk management.

- follow-up required for breaches of limits;
- (f) reviewing information on the Bank's liquidity risk developments on an ongoing basis, including the composition and characteristics of assets and funding sources, as well as market trends and events that may present significant or unprecedented challenges. This enables timely adjustments to the liquidity and funding strategy in response to changing environments and material developments;
 - (g) regularly reporting to the board on key risks and developments. Where there are new or emerging liquidity concerns, senior management should inform the board in a timely manner;
 - (h) implementing adequate internal controls to maintain the integrity of the liquidity risk management process. This includes:
 - adequate separation of risk-taking (i.e. first line of defence) and risk control (i.e. second line of defence) responsibilities;
 - having operationally independent, appropriately trained, and competent personnel to be responsible for implementing internal controls, with the critical ability to challenge information and modelling assumptions provided by business lines/front office;
 - sufficient oversight of the risk-taking function by an independent risk control function or functions that are well-resourced and have adequate stature and visibility within the Bank;
 - regular reviews by an independent party/internal audit (i.e. third line of defence) on the various components of the Bank's liquidity risk management process. This includes assessing the extent to which the Bank's liquidity risk management meets the expectations in supervisory guidance and industry sound practices, considering the Bank's liquidity risk profile and complexity, and examining the implementation and effectiveness of the agreed framework for controlling liquidity risk;
 - (i) ensuring that the new product approval process incorporates robust liquidity risk assessment, including identification and evaluation of potential liquidity risk exposures and the establishment of appropriate mitigants, before launching any new products or activities;
 - (j) ensuring that the Bank's analytical framework for measuring and attributing liquidity costs, benefits, and risks is reviewed regularly to reflect changing business and financial market conditions, thereby maintaining appropriate alignment of incentives;
 - (k) overseeing the implementation of the stress testing process, regularly reviewing scenarios, assumptions and results, and using stress tests to inform liquidity management;
 - (l) reviewing and updating the Bank's contingency funding plan at least annually or more frequently when necessary; and

(m) establishing adequate training programmes to equip staff to develop a thorough understanding of their roles and responsibilities, gain familiarity with the Bank's liquidity risk management processes and crisis protocols to be able to respond effectively during stress events. Training materials should be regularly updated to incorporate emerging risks, industry developments and insights from industry stress events and internal exercises.

- 3.4 A Bank should clearly identify the individuals or committees responsible for implementing and making liquidity risk decisions. While the board may delegate some of its liquidity risk management responsibilities to the Bank's Asset and Liability Committee (ALCO) or similar committees, it remains accountable and cannot abrogate its overall responsibility for the Bank. The board should maintain active oversight of these committees. For effective liquidity risk governance, the ALCO or similar committees should have a diverse membership comprising senior personnel from the treasury, risk management and key business units whose activities impact the Bank's liquidity risk profile. By bringing together all critical decision-makers and stakeholders, the committee can better evaluate the Bank's liquidity profile and potential risks, and implement effective liquidity management strategies.
- 3.5 The board and senior management should have an adequate understanding of potential interactions between liquidity risk and other risk types (e.g., asset market liquidity, credit, market, operational, legal, reputational, strategic and environmental) and consider these interactions in their decision-making processes.

4 Risk Appetite

- 4.1 A Bank should set its liquidity risk appetite to reflect the level of liquidity risk it is willing to assume. The appetite should be appropriate for the Bank's business strategy and its role in the financial system, and should reflect the Bank's overall financial condition and funding capacity. For instance, a Bank that is systemically important to the financial system would be expected to incorporate additional prudence when establishing its risk appetite.
- 4.2 A Bank's risk appetite should cover all key elements of liquidity risk it faces. Where material to the Bank's risk profile, this would include intraday liquidity risk and currency mismatch liquidity risk. The risk appetite should be sufficiently prudent to ensure that the Bank is able to withstand a range of stress scenarios, including a severe and rapidly escalating crisis and a prolonged period of stress.

- 4.3 A Bank's risk appetite should be clearly documented and understood by all relevant levels of management in order to impact decision-making. This includes understanding the trade-offs between risks and profits. A Bank may choose how best to structure, express and label its risk appetite measures. A combination of qualitative descriptions and quantitative measures is typical. For example, a Bank may outline its guiding principles for managing liquidity risk and specify a quantitative risk appetite measure, such as a minimum survival period under a set of severe but plausible stress scenarios⁷.
- 4.4 To facilitate adherence to the risk appetite, a Bank should establish additional operational limits, such as limits on cash outflows over various horizons, concentration of funding sources, unsecured short term wholesale funding, minimum level and composition of liquid assets, limits on acceptable levels of currency mismatches and encumbrance of assets.

5 Strategy and Incentives

- 5.1 A Bank should establish a liquidity management strategy which sets out its general approach to liquidity risk. The strategy should take account of liquidity needs in both normal and stressed conditions (considering institution-specific, market-wide, and combined scenarios). The strategy should align with the Bank's business profile and risk appetite, and be updated as market and macroeconomic conditions shift. It should also be appropriate for the nature, size and complexity of a Bank's activities. When formulating its strategy, the Bank should consider its legal structures, key business lines, the breadth and diversity of markets, products, jurisdictions in which it operates, and home and host regulatory requirements. The Bank should identify risks associated with its adopted strategy, and build in ways to mitigate the risks and factor in potential second-round effects from planned actions, where relevant.
- 5.2 The liquidity management strategy should be clearly documented. It should include specific policies on liquidity risk management such as:
- (a) composition and maturity of assets and liabilities;
 - (b) diversity and stability of funding sources;

⁷ The minimum survival period is the duration during which a Bank must maintain a positive net cumulative cash flow position under specified stressed conditions, without resorting to extraordinary liquidity assistance from the MAS or other relevant central banks.

- (c) management of liquidity across currencies, borders, business lines and legal entities;
- (d) intraday liquidity management;
- (e) holding of liquid assets, including minimum targets and composition, assumptions on the liquidity and marketability of assets considering market depth in stress; and
- (f) funding in relation to business targets and protection of franchise under stress.

The strategy may include various high-level quantitative and qualitative targets.

- 5.3 To implement its strategy, a Bank should maintain a funding plan that incorporates essential components, including forecasted balance sheets, projected funding needs and sources, and projected liquidity metrics, alongside other relevant factors. The plan should be regularly reviewed and updated to reflect changes in both the Bank's strategy and market conditions.
- 5.4 A Bank should measure and incorporate liquidity costs, benefits and risks in its internal pricing, performance measurement and new product approval processes for all significant activities⁸. This will align the risk-taking incentives of individual business lines with the liquidity risk exposures their activities create for the Bank as a whole.
- 5.5 Explicitly attributing the liquidity costs, benefits and risks to the relevant business activities (e.g., positions, portfolios or individual transactions) helps to ensure that line management incentives are consistent with and reinforce the overarching liquidity risk appetite and strategy of the Bank. This attribution should incorporate factors related to the anticipated holding periods of assets and liabilities, their market liquidity risk characteristics, and any other relevant factors, including the benefits from having access to relatively stable sources of funding, such as some types of retail deposits.
- 5.6 The quantification and attribution of liquidity risks should be explicit and transparent at the line management level and should include consideration of how liquidity would be affected under stressed conditions.

6 Policies and Procedures

- 6.1 A Bank should establish comprehensive liquidity risk policies and procedures that are aligned

⁸ These include both on- and off-balance sheet activities (which involve the creation of contingent exposures, which may not immediately have a direct balance sheet impact).

with its liquidity risk appetite and strategy, and that clearly define its liquidity risk management framework and operational processes. The scope and detail of these policies and procedures should be commensurate with the nature, size and complexity of the Bank's activities. These policies and procedures should encompass the following areas, as relevant to the Bank:

- (a) roles and responsibilities of the board, senior management and other personnel responsible for managing liquidity risk, including clear lines of authority for liquidity management decisions;
- (b) liquidity risk management structure for managing and overseeing liquidity positions of legal entities, branches and subsidiaries (if applicable);
- (c) liquidity risk management structure for reporting and coordination with head office, parent or group (if applicable);
- (d) liquidity risk management framework, tools and processes covering topics such as:
 - liquidity risk measurement methodologies, including methodologies for projecting cash flows over various time horizons;
 - liquidity risk controls, encompassing limits and thresholds, management action triggers, monitoring tools and early warning triggers;
 - liquidity risk management across individual currencies, business lines, legal entities, jurisdictions and at a group-wide level⁹;
 - intraday liquidity risk management;
 - liquidity risk management systems, monitoring and reporting;
 - framework for assigning costs and benefits related to internal use and provision of liquidity;
 - maintenance of liquid assets;
 - asset encumbrance;
 - collateral management;
 - stress testing and scenario analysis; and
 - contingency funding plan;
- (e) approval authority, review requirements and processes for managing limit breaches, policy deviations, and exceptions, including escalation procedures.

6.2 Policies should be properly documented, approved, and communicated across the organisation.

⁹ Where relevant, the policies should address how the group/head office/parent exercises monitoring and oversight over the liquidity positions of the individual entities, foreign branches, and subsidiaries as well as roles and responsibilities for managing liquidity risk in local and foreign currencies.

Supporting procedures should be established to implement risk management policies effectively through controls, checks and monitoring mechanisms. These procedures should be documented in sufficient detail to ensure proper execution. Policies and procedures should be reviewed regularly (e.g., annually) and updated as needed to reflect changing circumstances. A Bank should justify its policy review frequency and how the chosen frequency is sufficient to ensure that its policies remain current, relevant and effective in managing liquidity risk for its operations.

- 6.3 All business units conducting activities that have an impact on the Bank's overall liquidity should be fully aware of the liquidity risk appetite and strategy and operate under the approved policies, procedures, limits and controls.
- 6.4 The internal audit function should regularly review the effectiveness and implementation of the approved liquidity risk management framework, policies and procedures, providing independent assurance to the board and senior management.
- 6.5 While a foreign bank branch may manage liquidity risk globally as part of head office's group-wide liquidity risk management framework, it is expected to maintain policies and procedures to govern the Singapore branch's operations. These can be based off head office framework and adapted to the Singapore context where necessary to meet the expectations in the Guidelines.

7 Risk Identification

- 7.1 A Bank should define and identify the liquidity risk exposures for all legal entities, branches and subsidiaries in the jurisdictions in which it is active. As a Bank's liquidity needs and the sources of liquidity depend on its business and product mix, balance sheet structure and cash flow profiles of its on- and off-balance sheet obligations, it should evaluate each major on- and off-balance sheet position/source and determine how these can affect liquidity risk. Regular reviews are essential to ensure comprehensive risk capture, accounting for new products or activities and changes in the operating environment that may introduce new risk drivers.

Good practices observed in comprehensively identifying liquidity risk drivers

- To introduce rigour to the risk identification process, some Banks set up an internal risk inventory, assessing which risks might arise and how they were being captured in their individual liquidity risk management framework, including whether each risk

driver was subject to measurement and monitoring, limit control and/or included in stress test analyses.

- These Banks identified various liquidity risk drivers associated with specific products and activities. Taking deposits as an example, risks identified included runoff risks, rollover risks, early termination risks, and concentration risks across counterparties, maturities, and products. Specific factors that were found to affect outflows (e.g., counterparty type, business type and deposit balance tier) were flagged in the inventory to check if they were being considered in liquidity stress tests.
- For derivatives, these Banks identified and evaluated relevant risk drivers of potential outflows, including increases in initial margin requirements and collateral haircut, changes to variation margin from a change in market value of derivative contracts, downgrade trigger risk, and the risk of counterparties exercising their rights to substitute collateral of high quality with collateral of lower quality, and recall sleeper collateral. They then developed models/methodologies to quantify potential outflows for each applicable liquidity risk driver/component.

8 Risk Measurement and Evaluation

- 8.1 A Bank should have a sound process for accurately measuring liquidity risk. Given that no single metric can fully quantify liquidity risk, it is important for a Bank to use a range of metrics for measurement.
- 8.2 A Bank with simpler business models and less complex operations may adopt simpler measurement methodologies, provided these adequately capture its material liquidity risks. The sophistication of measurement tools and the range of metrics employed should be commensurate with the nature, size and complexity of the Bank's activities.

Cash Flow Projection

- 8.3 A Bank should implement a robust framework to comprehensively project cash flows arising from assets, liabilities, off-balance sheet items and derivatives over appropriate time horizons. A forward-looking cash flow analysis is a key tool used in liquidity risk management, providing the foundation for liquidity planning. It involves assessing a Bank's cash inflows against its

outflows and the liquidity value of its assets to identify the potential for future net funding shortfalls.

- 8.4 A Bank should forecast cash flows under both normal conditions and across a range of stress scenarios, including severe stress. For business-as-usual scenarios, the analysis would identify potential requirements that may arise from projected outflows relative to routine sources of funding under normal conditions. For stress scenarios, the analysis should identify funding requirements under stressed conditions. The Bank should factor in vulnerabilities to potential events, activities and strategies that can significantly strain internal cash generation capability.
- 8.5 Cash flow projections should allow stakeholders to assess vulnerabilities to changes in liquidity needs and funding capacity across multiple time horizons: intraday exposures, day-to-day liquidity demands and funding availability over short and medium-term periods extending to one year, and structural liquidity requirements beyond one year.
- 8.6 A Bank should make realistic assumptions about the behaviour of its assets, liabilities and off-balance sheet items when determining potential cash flows.¹⁰ This is particularly important as contractual maturities for certain products may be either unspecified or may not accurately reflect actual cash flow behaviour. Moreover, to develop prospective, dynamic cash flow forecasts, a Bank should incorporate assumptions about the likely behavioural responses of both the Bank itself and its key counterparties under various scenarios.
- 8.7 Such projection should be conducted at a sufficiently granular level for more accurate forecasting by taking into account specific factors affecting different types of cash flows. For this purpose, a Bank should determine the appropriate level of segmentation to calibrate and apply outflow and inflow assumptions, considering potential differences in customer behaviour and franchise needs. Adequate segmentation is essential for improving risk discovery as it better captures the risk characteristics of individual segments.
- 8.8 Cash flow projections and analyses should be comprehensive and incorporate, at a minimum, the following components:
 - (a) future cash flows of assets and liabilities;
 - (b) sources of contingent liquidity demand and related triggers associated with off-balance

¹⁰ Such assumptions may include behavioural patterns of assets and liabilities without defined maturity dates, probability of asset and liability rollovers and early termination, potential liquidity demands arising from off-balance sheet activities, and accessibility and capacity of various funding sources and asset markets.

- sheet positions;
- (c) correspondent, custody and settlement activities; and
- (d) currencies in which a Bank is active.

Future cash flows of assets and liabilities

- 8.9 When estimating outflows arising from its liabilities, a Bank should assess the “stickiness” of its funding sources, including the likelihood of funding roll-overs, particularly from large fund providers, under both business-as-usual and stress scenarios. For liabilities with embedded optionality where the timing and amount of withdrawals are uncertain, a Bank should conduct analyses to determine their cash flow patterns under both business-as-usual and stress scenarios. A Bank should also be mindful that liabilities that exhibit stable behaviour under business-as-usual conditions can experience materially higher outflows under stress. Please refer to section 11 for details on stress testing and scenario analysis.
- 8.10 To estimate inflows, a Bank should analyse the quality and marketability of assets that could be sold off or serve as collateral for secured funding. A Bank should ensure that assets are prudently valued in accordance with relevant financial reporting and supervisory standards. In projecting inflows from assets, a Bank should factor in the accessibility and capacity of various funding markets (across different channels, markets, currencies, instrument of funding) under different scenarios. It should assess how market stress affects both the value of assets and their marketability when evaluating the impact on its liquidity position. Please refer to paragraph 11.19 for details. The lead time required for the monetisation of assets should take into account the settlement periods and the impact of time zone differences where applicable. A Bank would also need to make assumptions regarding the extent to which matured assets are renewed and new assets are acquired.

Sources of contingent liquidity demand and related triggers associated with off-balance sheet positions

- 8.11 A Bank should identify and measure potential cash flows relating to off-balance sheet commitments and other contingent liabilities. A Bank’s processes for identifying and measuring contingent funding risks should consider the nature and size of the Bank’s potential non-contractual “obligations”, as such obligations can give rise to the Bank supporting related entities in times of stress.
- 8.12 A Bank should establish policies and procedures to identify, assess and manage step-in risk as

part of its broader liquidity risk management framework.¹¹ This risk can manifest in terms of providing support to off-balance sheet vehicles, particularly securitisation and conduit programmes, where the Bank considers such support critical to maintaining ongoing access to funding; supporting unconsolidated entities with which the Bank has sponsorship, investment, or other relationships that could expose it to reputational risk; and purchasing assets from money market or other investment funds that the Bank manages or with which it is otherwise affiliated for reputational reasons.

8.13 Given their prevalence and potential challenges in assessing the related liquidity risks that could materialise in times of stress, the following off-balance sheet items warrant particular attention.

- (a) Special Purpose Vehicles (SPVs): A Bank should have a thorough understanding of its contingent liquidity risk exposure and event triggers arising from any contractual and non-contractual relationships with SPVs. Where a Bank provides contractual liquidity facilities to an SPV or is expected to support the SPV's liquidity under adverse conditions, the Bank needs to consider how its liquidity might be adversely affected by illiquidity at the SPV. In such cases, the Bank should monitor the SPV's inflows (maturing assets) and outflows (maturing liabilities) as part of the Bank's own liquidity planning, including in its stress testing and scenario analyses. The Bank should assess its liquidity position with only the SPV's liquidity draws included but not its liquidity surplus.¹² If a Bank has received a deposit of surplus cash from an SPV, it should consider the potential for withdrawal of such deposits. With respect to the use of securitisation SPVs as a source of funding, a Bank needs to evaluate whether these funding vehicles will continue to be available under adverse scenarios. A Bank experiencing adverse liquidity conditions often will not have continuing access to the securitisation market as a funding source and should reflect this in its liquidity management planning.
- (b) Financial Derivatives: When managing liquidity risks associated with financial derivatives, a Bank should incorporate cash flows related to the repricing, exercise and/or maturity of financial derivatives contracts in its liquidity risk analysis, including associated margin

¹¹ Banks may refer to the Guidelines on Identification and Management of Step-in Risk issued by the BCBS on 25 October 2017 for guidance (<https://www.bis.org/bcbs/publ/d423.htm>). Step-in risk refers to the risk that a Bank will provide financial support to an entity beyond, or in the absence of, its contractual obligations should the entity experience financial stress.

¹² The SPV's liquidity surplus should not be included as a source of liquidity under adverse conditions, because: i) when a Bank is experiencing severe strain, the SPV's cash surplus may cease to be available to the Bank (e.g., the SPV's managers may decrease exposure to the Bank by depositing funds with another bank); and ii) a high correlation often exists between liquidity strains for most Banks and the SPVs they sponsor and administer (e.g., concerns related to a Bank's financial strength or the SPV's performance can trigger liquidity pressures for the other entity).

requirements.

- (c) Guarantees and Commitments: Undrawn loan commitments, letters of credit and financial guarantees represent potentially significant outflows of funds for a Bank. A Bank should assess these potential outflows using the most appropriate method. For instance, a Bank may ascertain a "normal" level of cash outflows under business-as-usual conditions, and then estimate the scope for an increase in these outflows during periods of stress. It should consider the nature of the commitment and credit worthiness of the counterparty, as well as exposure to industry sectors and geographical locations, as there can be correlation and amplification in stresses experienced across industry sectors and/or geographical locations. The customised nature of contracts underlying undrawn commitments and off-balance sheet instruments can pose challenges in modelling triggering events¹³ for contingent liquidity risks. A Bank should implement systems and tools to analyse liquidity triggering events effectively and to measure the potential impact of underlying risk factor changes on draws against these facilities, even if there has been no past experience of such draws. This analysis should include appropriate assumptions on the behaviour of both the Bank and its obligors or counterparties.

Correspondent, custody and settlement activities

- 8.14 A Bank should understand and have the capacity to manage how the provision of correspondent, custodian and settlement bank services can affect its cash flows. Given that the gross value of customers' payment traffic (inflows and outflows) can be very large, unexpected changes in these flows can result in large net deposits, withdrawals or line of credit drawdowns that impact the overall liquidity position of the correspondent or custodian bank, both on an intraday and overnight basis. A Bank should also understand and have the capacity to manage potential liquidity needs it would face as a result of the failure-to-settle procedures of payment and settlement systems in which it is a direct participant.

¹³ Triggering events are events which enable commitments to be drawn upon and thus may create a liquidity need. For example, triggering events could include changes in economic variables or conditions, credit rating downgrades, country risk issues, specific market disruptions (e.g., commercial paper), and the alteration of contracts by governing legal, accounting, or tax systems and other similar changes.

Currencies in which a Bank is active¹⁴

- 8.15 In addition to forecasting cash flows on an all-currency basis, a Bank should measure and perform a separate analysis of liquidity risk for each significant currency. Please refer to section 14 for details.

Cash flow projection assumptions

- 8.16 A Bank should take steps to ensure that the assumptions it uses for cash flow projections are reasonable and appropriate. Such assumptions, including the underlying considerations, should be clearly documented and periodically reviewed and approved. Where a Bank utilises models to measure liquidity risks, the development and use of models should be subject to the Bank's internal model risk management policy standards and procedures, such as assessment and assignment of individual model risk ratings, independent model validation, regular model risk reviews, back-testing and model performance monitoring. A Bank should also deploy adequate processes and control mechanisms to ensure the quality of data used in the modelling.
- 8.17 A Bank should assess coherence between behavioural assumptions used in liquidity risk management and those applied in Interest Rate Risk in the Banking Book (IRRBB). While MAS recognises that these frameworks serve different risk management objectives and may appropriately employ different assumptions and scenarios, there should be some coherence in how the underlying behavioural characteristics of customer segments are understood and described across frameworks. A Bank should be able to justify any significant divergence in behavioural assumptions between its liquidity risk and IRRBB frameworks, particularly where such differences could understate potential risks.¹⁵
- 8.18 For business-as-usual analyses, a foreign bank branch may use contractual cash flow projections without applying behavioural assumptions if it assesses that contractual cash flow projections adequately capture or are appropriately conservative for its risk profile. One example could be a foreign bank branch that is primarily funded by wholesale funding.

¹⁴ This generally refers to a currency where the aggregate liabilities of a Bank denominated in that currency amounts to 5% or more of its total liabilities. Nevertheless, a Bank should assess whether currencies below this threshold may be significant based on other relevant factors. For example, the operating currency of a strategically important subsidiary within a banking group may be considered significant even if its exposure is less than 5%.

¹⁵ For the avoidance of doubt, a Bank is not expected to directly map the assumptions used for its liquidity management with its IRRBB risk management.

Other Risk Measures

8.19 To gain a comprehensive picture of its liquidity position and profile, a Bank should complement the use of cash flow analyses with other risk measures, appropriate to its business model and complexity, such as:

- (a) loans to deposits ratio, which measures the extent of coverage of loans by deposits, as a rough gauge of a Bank's liquidity, (e.g., whether it can self-fund its loans with customer deposits);
- (b) contractual maturity mismatch analyses, which identify the gaps between assets and liabilities, including off-balance sheet items, along maturity-defined time bands. These provide insights into the extent to which a Bank relies on maturity transformation and potential gaps that need to be funded in future;
- (c) funding source analyses, which track the extent of a Bank's funding provided by categories such as retail deposits, uninsured deposits, and wholesale funding, including their respective sub-categories;
- (d) concentration risk analyses, which measure funding from specific sources (e.g., top depositors, related group of fund providers, economic sector, geography, instrument type, secured versus unsecured market funding, currency and maturity, sources of new deposits) as a proportion of total funding;
- (e) currency mismatch analyses, which measure the funding gap in individual currencies and/or usage of swaps in funding;
- (f) intragroup lending and borrowing, which measures the amount of borrowing and lending within a group;
- (g) available unencumbered assets analyses, which provide information on assets that might be used as collateral to raise liquidity in secondary markets or are eligible for central bank facilities (size, composition, characteristics and location held);
- (h) asset encumbrance ratio, which measures the proportion of encumbered assets to total assets, indicating the extent to which assets are encumbered and the potential capacity for additional asset pledging to obtain funding. This metric is complemented by detailed information on encumbered assets, including their composition by maturity, asset type and source of encumbrance;
- (i) composition of liquid assets by accounting classification, including unrealised losses from held-to-collect debt securities that would be crystallised when sold, which provides insight into the potential impact on a Bank's capital position in the event such positions are sold.
- (j) liquid asset monetisation results, which provide information on a Bank's ability to convert its liquid assets into cash quickly and efficiently;



- (k) intraday liquidity monitoring tools, which provide insights into a Bank's intraday liquidity conditions and patterns. Please refer to paragraph 15.7 for examples of intraday liquidity monitoring tools;
- (l) undrawn commitments ratio, which measures the level of total undrawn commitments over total lending, giving an indication of potential funding needs;
- (m) stress test results, which estimate a Bank's liquidity positions under stress scenarios, including how long it can survive, or level of liquidity position post stress; and
- (n) regulatory metrics, which include Liquidity Coverage Ratio (LCR)/Minimum Liquid Assets Ratio (MLA), Minimum Cash Balance Ratio (MCB) and Net Stable Funding Ratio (NSFR).

- 8.20 A Bank should compute key liquidity risk measures at the levels of both all-currency and each significant currency.
- 8.21 Banking groups should consider local regulatory requirements and/or potential impediments to funds flow when computing the available liquidity to meet their needs within the group and in specific entities and/or locations, especially if liquidity might not be fully transferrable between entities within a group during stress scenarios.

9 Management Information System and Monitoring

- 9.1 A Bank should maintain effective and reliable monitoring systems and processes to provide the board, senior management and other relevant personnel with timely and forward-looking information on its liquidity position. The extent and sophistication of these capabilities should be commensurate with the nature, size and complexity of the Bank's activities and its exposure to rapid liquidity outflows. A Bank's management information system (MIS) should capture all sources of liquidity risk, including contingent risks and the related triggers. When a Bank introduces new products or activities, the impact on liquidity risk, if any, due to new cash flow implications should be likewise captured by the MIS.
- 9.2 The MIS of a locally-incorporated Bank that has a subsidiary/branch should have the ability to calculate liquidity positions in each of the currencies in which it conducts business, both at the group and subsidiary/branch levels.
- 9.3 To effectively manage and monitor its net funding requirements, a Bank should have the ability to calculate liquidity positions on an intraday basis, on a day-to-day basis for the shorter time

horizons (with daily cumulative cash flow monitoring), and over progressively wider time bands for longer-term periods. For short time horizons, daily cash flow monitoring is crucial as it helps identify peak liquidity needs that might occur between specified tenor bucket points which could otherwise be missed if only time-band analysis is used. For longer-term horizons, wider time bands are acceptable since there is more time available for implementing corrective measures and these positions will naturally transition into daily monitoring as they move closer to the present time horizon.

- 9.4 The system should be capable of delivering more granular and time sensitive information during stress events. This includes the ability to produce ad hoc liquidity reports to apprise the Bank's board, senior management, and MAS when required, of the latest liquidity positions.
- 9.5 Building on the MIS, a Bank should regularly monitor various liquidity risk metrics in compliance with its established policies, procedures and limits.
- 9.6 A Bank should establish and monitor early warning indicators (EWIs). These provide insights into the Bank's risk profile and flag potential deterioration in its liquidity position. A Bank should implement formalised processes to monitor these indicators, including clearly defined triggers or thresholds and escalation and reporting procedures. Frequency of monitoring and parties responsible for monitoring should also be clearly laid out. When indicators suggest increased risk, prompt assessment and, if necessary, management action should be taken to mitigate emerging risks.
- 9.7 A Bank should select and monitor EWIs that are relevant to its specific business model, risk profile and operations. EWIs can be qualitative or quantitative and may include, but are not limited to, the following:

Relating to a Bank's liquidity profile

- (a) rapid asset growth, especially when funded by unstable funding sources, which could lead to asset and liability mismatches;
- (b) rapid deposit growth, which could be vulnerable to sudden reversals;
- (c) growing concentrations in assets or liabilities (e.g., concentration to a specific sector or single borrower/depositor);
- (d) increases in currency mismatches;
- (e) repeated incidents of positions approaching or breaching internal limits;
- (f) deterioration in regulatory metrics (e.g., LCR, NSFR);

- (g) increasing retail and wholesale deposit outflows (e.g., increasing outflows from high value retail deposits and wholesale fund providers);
- (h) a decrease of weighted average maturity of liabilities;
- (i) increasing redemptions of certificates of deposit before maturity;
- (j) difficulty accessing longer-term funding;
- (k) difficulty placing short-term liabilities (e.g., commercial paper);
- (l) worsening intraday liquidity indicators (e.g., nostro accounts balances falling below the Bank's pre-set internal threshold, fast depletion of collateral account used to support real-time settlement channel, operational problems in payment and settlement systems);
- (m) deterioration in internal stress test results;

Relating to a Bank's financial conditions

- (n) significant deterioration in the Bank's earnings, asset quality, and overall financial condition;
- (o) negative trends or heightened risk associated with a particular product line, such as rising delinquencies;

Relating to counterparties' responses

- (p) counterparties request increased margins and additional collateral for credit granted or resist entering into new transactions;
- (q) correspondent banks eliminate or decrease the Bank's credit lines;

Relating to market indicators, funding costs and negative publicity on a Bank

- (r) rating agency credit rating downgrades or credit watches for potential downgrades;
- (s) stock price declines;
- (t) widening credit default swap spreads;
- (u) rising wholesale or retail funding costs in a stable market;
- (v) negative publicity;

Relating to market indicators that inform on general funding conditions

- (w) widening term funding – overnight indexed swap spreads; and
- (x) widening cross currency swaps spreads.

9.8 A Bank should also have EWIs that signal whether embedded triggers in certain products (e.g., callable public debt, over-the-counter derivative transactions) are about to be breached or whether contingent risks are likely to crystallise and cause the Bank to provide additional

liquidity support for the product or bring assets onto the balance sheet.

9.9 With the proliferation of digital banking, customers can now transfer funds instantaneously without the friction of requiring in-person withdrawals. Coupled with the speed of information (and misinformation) dissemination through social media, a Bank is more vulnerable to rapid deposit outflows when customer confidence is impacted. To address these developments and enable swift responses, a Bank should:

- (a) strengthen its MIS capabilities and tools to allow rapid assessment of its liquidity positions (e.g., in real-time), including beyond normal business hours; and
- (b) enhance its monitoring capabilities to analyse trends from both news and social media platforms, enabling timely intervention when necessary.

Good practices observed in news and social media monitoring

- Some Banks implemented comprehensive monitoring frameworks to monitor across multiple news and social media platforms. Both quantitative and keyword-based thresholds were set to pick up any negative mentions of the Banks. Some Banks also set more stringent thresholds for liquidity-related alerts.
- Some Banks were able to perform monitoring on a real-time basis, with customised alerts being generated within and outside of business hours.
- Dedicated teams within the Banks were identified to monitor and analyse news and social media chatter and to escalate to the appropriate stakeholders for follow-up actions, where necessary. These Banks also put in place process workflows for the monitoring and escalation of news and social media chatter, depending on a number of factors, such as the nature of the issues, traction on news/social media platforms, materiality of impact to the Bank and whether a coordinated group-wide response would be required.

10 Control and Reporting

10.1 Consistent with its established risk appetite, a Bank should set limits to control and manage its liquidity risk exposure. Such limits should be relevant to the Bank, taking into consideration the type and complexity of business activities, locations, products, currencies, markets served and

potential speed of deterioration. Please refer to section 8 for a list of risk metrics on which limits/triggers can be set to control the level of risk-taking.

- 10.2 The limits and the corresponding escalation procedures should be reviewed on a regular basis. There should be clear procedures on the investigation of non-compliances with the limits and the remediation actions to be taken to prevent such incidents from recurring. Breaches in liquidity risk limits should be reported, according to specified thresholds and reporting guidelines, to higher levels of management, the board and MAS where relevant.
- 10.3 A Bank's board should be made aware of the Bank's liquidity position each time it meets, and more frequently in times of stress. Reporting of risk measures should be done on a timely basis (e.g., real-time/intraday/daily depending on the metric) for those responsible for managing liquidity risk.

11 Stress Testing and Scenario Analysis

Framework

- 11.1 A Bank should be prepared to manage liquidity under stressed conditions. A Bank should thus perform stress tests or scenario analyses regularly using a variety of short-term (including intraday) and protracted institution-specific and market-wide stress scenarios (individually and in combination). These analyses quantify potential impacts on the Bank's cash flows, liquidity position, profitability and solvency to help identify potential areas of liquidity strain. For a locally-incorporated Bank, stress tests should be conducted at individual legal entity, overseas branch and group (or sub-group, if relevant) levels.¹⁶
- 11.2 The extent and frequency of a Bank's stress testing should be commensurate with its nature, size and complexity, as well as with the relative importance of the Bank within the financial system. A Bank should document its policy on stress testing frequency that reflects its risk profile. A Bank should have the capability to increase the frequency of stress tests and conduct ad hoc stress tests when warranted, such as during volatile market conditions or at MAS' request.
- 11.3 Stress tests should be performed for all currencies in aggregate as well as separately for each

¹⁶ Where relevant, stress test analysis at business lines should also be conducted.

significant currency.

- 11.4 To identify and analyse factors that could have a significant impact on its liquidity profile, a Bank may conduct an analysis of the sensitivity of stress test results to certain key assumptions. Such sensitivity analyses can provide additional indications of a Bank's degree of vulnerability to certain factors.
- 11.5 To explore tail risk, a Bank should consider using reverse stress tests by choosing a known stress test outcome (e.g., illiquidity) and then identifying the events that could result in such an outcome. Reverse stress tests enable a Bank to consider what tail events entail, assess their likelihood, and then decide on the actions to take.
- 11.6 A foreign bank branch may leverage its head office stress testing framework if it is able to demonstrate that the framework adequately captures liquidity risk exposures that are specific to its Singapore operations. Should the foreign bank branch rely on stress tests performed by head office, Singapore branch-level results should be promptly shared with local management. The Singapore branch management remains responsible for understanding and acting upon the stress test results that have a material impact on the Singapore operations.

Scenarios

- 11.7 When designing stress scenarios, a Bank should consider the nature of its business, activities and vulnerabilities so that the scenarios incorporate the major funding and market liquidity risks it is exposed to.¹⁷ These include risks associated with its business, activities, products (including complex financial instruments and off-balance sheet items), funding sources and liquidity demands (including potential concentrations).
- 11.8 Stress tests should cover the following types of severe but plausible scenarios.
 - (a) An institution-specific stress scenario would involve idiosyncratic events affecting just the Bank. This could lead to, among other things, significant deposit outflow, severe constraints in accessing wholesale funding markets and foreign exchange swap markets, and/or potential increases in collateral requirements.

¹⁷ A Bank should leverage on its liquidity risk identification and assessment processes to ensure comprehensive consideration and capture of liquidity risk drivers in stress tests with these assessments clearly documented and regularly reviewed.

- (b) A market-wide stress scenario would consider the systemic dimension of liquidity crises. This might include the simultaneous drying up of all or most of the key funding markets including those previously highly liquid markets, herd behaviour, interaction between reduction in market liquidity and constraints on funding liquidity, and/or significant market volatility leading to margin calls. Other scenarios might be disruptions in foreign exchange swap markets and/or restrictions on currency convertibility, severe operational or settlement disruptions affecting one or more payment or settlement systems, second-round effects due to behavioural interactions among market participants and funding needs related to off-balance sheet vehicles and exposures.
- (c) A combined stress scenario means that both institution-specific and market-wide stress events are occurring simultaneously. A Bank might be hit with a credit downgrade at a time when there is general tightness in the market due to other factors and its borrowers are unable to make timely repayments. Sharp drops in asset prices and increases in price volatility could lead to higher haircuts on collateral. There could also be larger drawdowns of credit lines by customers and margin calls on a Bank, and/or a disinclination of market participants to trade in over-the-counter swap markets amidst seize-up in the markets.

11.9 Given the uncertainty surrounding the duration of potential liquidity stresses and for risk discovery purposes, a Bank should perform stress tests across a range of time horizons. These include both short-term (including intraday) and protracted periods of liquidity stresses. They can encompass rapidly escalating scenarios¹⁸ and more prolonged situations where the Bank's liquidity position gradually deteriorates. A Bank should assess the appropriateness of stress horizons used given its balance sheet profile, strategy, business model and activities.

11.10 It is important for a Bank to consider the insights and results of stress tests performed for various other risk types when stress testing its liquidity position and evaluate possible interactions between liquidity risks and other risks. For instance, stress test scenarios should consider how losses and the resulting reduction in capital can impact a Bank's ability to maintain funding relationships. Similarly, the outcome of liquidity stress tests should feed into the stress tests for other risk types where relevant.

¹⁸ Such as one with front-loaded, severe deposit outflows in the early days of a stress to account for potential digitalisation-enabled run dynamics.

Assumptions

- 11.11 A Bank should have a sound basis for stress assumptions. In setting assumptions, a Bank can take reference from (a) its own experience in past stresses factoring in risk drivers from changes in balance sheet and risk environment; and (b) past stress events experienced by other banks, accounting for differences in profiles. When developing stress test assumptions, a Bank should err on the side of conservatism where there is significant uncertainty about cashflow behaviour under stress. For stress testing to be forward-looking, these assumptions should account for ongoing changes in a Bank's strategy, business activities and risk profile, changes in the markets in which it operates, and new developments that may not be covered by historical stress events. Assumptions also need to be made about the capacity of various funding markets and conditions of asset markets during stress.
- 11.12 Stress test assumptions should be determined and applied at a reasonably granular level to allow for robust assessment of vulnerabilities in outflows, inflows and actions to generate liquidity. The following assumptions relate to the behaviour of contractual and non-contractual cash flows arising from on- and off-balance sheet items and other contingent liabilities.

On-balance sheet

- 11.13 For liabilities without contractual maturity, a Bank should estimate the run-off behaviour (amount and speed) under stress. Additionally, for term liabilities with embedded options for early withdrawal, a Bank should take into account the potential for early withdrawal under stress scenarios. For liabilities with contractual maturities including secured funding, a Bank should assess whether its ability to roll over funding under stress is reduced.
- 11.14 In all these assessments, where relevant, a Bank should factor in how the availability of digital banking channels would potentially lead to larger and faster outflows than traditionally assumed. It is prudent for a Bank to calibrate its stress test assumptions accordingly to reflect this reality.
- 11.15 Different types of deposits/funding may exhibit varying levels of "stickiness" or potential speed of outflows. Factors to consider include the size of deposits, deposit insurance coverage, nature of deposits (e.g., operational versus non-operational), method of obtaining funding (e.g., promotional campaigns with higher interest rates), deposit channel (e.g., direct internet or brokered), and interest rate sensitivity. The stability of funding under stress can also be influenced by the characteristics of fund providers, for instance, their industry sector (e.g.,

whether they are financial or non-financial institutions) and geographical location (which may affect their behaviour during regional or global market stress).

- 11.16 A Bank should evaluate the likelihood of fund providers behaving similarly under stress, and the possibility of both secured and unsecured funding drying up. For secured funding with overnight maturity, a Bank should not assume that the funding will automatically roll over. A Bank should also assess the availability of term funding backing up facilities and the circumstances under which they can be utilised.
- 11.17 Stress test assumptions should appropriately consider the potential impact of concentration within a Bank's balance sheet on its liquidity under stress. For instance, a Bank may have concentration in its funding profile (e.g., by individual, related depositors, geography location, industry), assets (e.g., liquid assets portfolios), and/or off-balance sheet positions (e.g., concentrated loan commitments). The materialisation of any concentration risk factor could significantly impact the Bank's liquidity position and should therefore be carefully assessed in stress scenarios.
- 11.18 When making assumptions about inflows from maturing loans, due consideration should be given to the need to roll over existing loans for franchise reasons, or that borrowers may be affected from a market-wide stress and be unable to repay on time. The Bank may also need to continue granting new loans to stay in the market.
- 11.19 In estimating the market value of liquid assets during stress, a Bank should take into account potential price changes and valuation erosion under different stress scenarios. A liquidation cost analysis¹⁹ would be useful to assess the impact of liquidation on meeting immediate liquidity demands, rebuilding liquidity resources, or bringing risk exposure back within the liquidity risk appetite. This analysis should consider the size of its positions relative to market liquidity under stressed conditions and potential for simultaneous liquidation by multiple market participants in stressed conditions, estimate potential losses and cash generated from liquidation incorporating incremental market impact and liquidation costs in view of reduced market depth and wider bid-ask spreads, and evaluate the portfolio's liquidity profile post liquidation.
- 11.20 Assumptions should reflect accurate timeframes for the settlement cycles of assets that might be liquidated, and the time required to transfer liquidity across borders. In addition, if a Bank

¹⁹ For additional guidance on this topic, a Bank may refer to the Liquidity Preparedness for Margin and Collateral Calls: Final Report issued by the Financial Stability Board on 10 December 2024 (<https://www.fsb.org/2024/12/liquidity-preparedness-for-margin-and-collateral-calls-final-report/>).

relies upon liquidity outflows from one payment system to meet obligations in another, it should consider the risk that operational or settlement disruptions might prevent or delay expected flows across systems. This is particularly relevant for firms relying on intragroup transfers or centralised liquidity management. Where relevant, a Bank should also consider the ability and cost to convert currencies under stressed market conditions, and the additional settlement time required to settle foreign exchange conversion.

Off-balance sheet

- 11.21 A Bank should evaluate the potential liquidity risks from off-balance sheet commitments, such as issued guarantees and undrawn credit lines (both committed and uncommitted) granted to third parties or intragroup. For instance, under a market-wide stress scenario, there could be an increase in the amount of drawdown of credit lines and guarantees. Counterparties may also request additional margin or collateral as a result of a downgrade of a Bank's credit rating, a decrease in the mark-to-market valuation of open derivative positions, or a change in the value of collateral that a Bank has provided.
- 11.22 Regarding asset sale and securitisation, a Bank should consider potential increases in liquidity demands arising from recourse provisions in asset sales, the extension of liquidity facilities to securitisation programmes, and early amortisation triggers of certain asset securitisation transactions. There could also be liquidity drains due to non-contractual obligations. Please also refer to paragraphs 8.12 and 8.13.
- 11.23 For committed lines of credit or guarantees provided by others, a Bank is expected to realistically assess whether or to what extent such facilities can be tapped on during a stress situation. Similarly, a Bank holding assets that are guaranteed by a third party or has used such assets as collateral for raising funds could face liquidity demands if the third party's credit standing is closely linked to the asset quality. The value of protection purchased from guarantors could decline when the asset value deteriorates, requiring a Bank to provide additional margin for borrowings against such assets.

Other considerations

- 11.24 A Bank should consider how other market participants might react to market stress (e.g., any change in risk appetite) and how this could amplify market movements or stress due to a common response. For instance, market participants may hoard liquidity and not on-lend in term interbank markets, leading to illiquidity spirals. If a Bank uses a correspondent or custodian for

settlement, the analysis should include the impact of those agents restricting their provision of intraday credit. It should also consider the likely impact of its own behaviour on that of other market participants.

- 11.25 A Bank that has cross-entity and/or cross-jurisdiction funding should evaluate the impact of potential disruptions to material cross-border funding channels and/or currencies as part of its stress testing. It should analyse how the liquidity positions of its group entities may impact its group liquidity, either directly or through contagion when those entities face liquidity strain. A Bank that is part of a group should consider the appropriate treatment of its intragroup transactions, including short-term funding and committed liquidity lines provided to, or received from, other group entities in a stress scenario. The appropriate approach depends on whether the stress scenario is localised or affects the entire group. In a localised stress scenario, a Bank should only include cash inflows from intragroup funding lines if the funding arrangement is fully committed and irrevocable, and there is a reasonable level of certainty that the funding will be received in stressful situations. In scenarios where stress affects the entire group, a Bank should carefully evaluate the availability and reliability of intragroup, head office or parent funding support. While such support might be available in some cases, its effectiveness could be significantly compromised when the entire group is under stress.
- 11.26 An illustrative list of risk drivers to be considered in a liquidity stress test include:
- (a) run-off of retail funding;
 - (b) unavailability/impaired unsecured/secured wholesale funding sources;
 - (c) increased correlation between funding markets reducing the benefits of diversification across sources of funding;
 - (d) shortening of funding tenors;
 - (e) increased margin calls and collateral requirements;
 - (f) increased contingent claims and draws on committed lines extended to third parties or the Bank's subsidiaries, branches or head office;
 - (g) increased liquidity demands from off-balance sheet vehicles and activities (including conduit financing);
 - (h) heightened liquidity drains associated with complex products/transactions;
 - (i) negative impact of credit rating downgrades;
 - (j) heightened risks arising from concentrated and leveraged positions;
 - (k) asset market illiquidity and erosion in the value of liquid assets;
 - (l) reduction or loss of contingent lines extended to the Bank;
 - (m) impaired foreign exchange convertibility and restricted access to foreign exchange markets

- under stress;
- (n) constraints on liquidity transfer across entities, sectors and borders taking into account legal, regulatory, operational and time zone restrictions and constraints;
- (o) restricted access to central bank facilities due to operational limitations;
- (p) impaired operational ability to monetise assets;
- (q) inadequate remedial actions and the lack of necessary documentation and operational expertise and experience to execute them, taking into account the potential reputational impact when executing these actions;
- (r) accelerated balance sheet growth;
- (s) early debt buyback calls from investors; and
- (t) surge in intraday liquidity demands.

A Bank should assess whether the risk drivers, including other risk drivers not listed, are material to its operations and focus its stress testing accordingly.

Review of Methodology and Assumptions

- 11.27 A Bank should regularly review its stress test scenarios and assumptions to ensure that the nature and severity of the tested scenarios remain appropriate for the Bank, and that they are able to surface the relevant risks and vulnerabilities. The review should take into account changes in market conditions, changes in the nature, size or complexity of a Bank's business model and activities and actual experiences in stress situations. The review process should be carried out by parties who have the appropriate expertise and were not directly involved in setting the said assumptions and methodology. The scope of review should include all models, methodologies and assumptions (including overlays and expert judgments) used in stress testing. The outcomes and basis of each review should be clearly documented.
- 11.28 In the absence of suitably qualified reviewers who were not involved in setting the assumptions and methodology due to a Bank's limited operation in Singapore, it is possible for a management oversight committee or the Bank's head office to perform the review, provided they have the appropriate expertise. In all cases, senior management is expected to oversee the stress testing process and exercise judgement in assessing the appropriateness of the scenarios and assumptions, as mentioned under paragraph 3.3(k) of the Guidelines. Separately, MAS expects a Bank's internal audit function to audit and provide assurance on the overall stress testing process as part of its independent oversight responsibilities.
- 11.29 Certain scenarios and assumptions may be developed centrally at the head office/parent/group

level before being implemented at its overseas locations. These should still be subject to review to determine their suitability for its Singapore operations. Additional local scenarios and assumptions should be run where warranted to reflect the unique local circumstances.

- 11.30 Singapore-headquartered banking groups should have a set of consistent scenarios and assumptions to run on a group-wide basis to assess the group’s resilience to various liquidity conditions. However, circumstances may likewise warrant differentiated treatment (e.g., a particular outflow assumption) to better reflect local conditions. These can be factored in after careful consideration by head office/parent/group.

Utilisation of Results

- 11.31 Stress test outcomes should inform a Bank’s liquidity risk management strategies, policies, and setting of limits and enable the development of effective contingency plans. Stress test results, together with key scenario assumptions and vulnerabilities, along with any recommended actions should be reported to and discussed with the board and senior management. To the extent that the stress test outcomes exceed a Bank’s liquidity risk appetite, senior management should adjust liquidity positions and/or bolster contingency plans in consultation with the board. The Bank should notify MAS of the results and rectification actions if the impacts are material.

Good practices observed in liquidity stress testing practices

Framework, governance and process

- Some Banks monitored daily cumulative stressed cash flows to identify mismatches within the stress test time horizon, whilst ensuring sufficient buffers were available to cover any cash flow gaps throughout the stress test time horizon. This helped to address cash flow mismatch risk within the relevant horizon.
- Some Banks conducted stress tests using scenarios that covered various time horizons. Some scenarios extended beyond the 30-day regulatory LCR horizon. This extended view helped Banks identify and manage potential “cliff effects” that could arise from maturity mismatches in their liquidity profiles, which might otherwise remain hidden when focussing solely on the standard 30-day window.
- Quarterly reviews of stress test assumptions were conducted by some Banks to assess appropriateness to incorporate market, macroeconomic, business and regulatory changes, including new products or activities.

Liquidity outflow and inflow assumptions

- Some Banks applied additional conservatism where stressed data was unavailable or there was great uncertainty about cashflow behaviour under stress. For example:
 - using the worst historical outflows as a starting point for calibration;
 - applying overlays to reflect hypothetical worst-case scenarios (e.g., scaling up historical worst outflow by assuming no government support or intervention to backstop outflows); and
 - assuming larger outflows in the beginning period of the stress scenario to test the ability to withstand sudden and rapid outflows under stress.
- Some Banks performed benchmarking using peer analysis and failed bank data during periods of stress to calibrate outflow rates and assess overall appropriateness of stress assumptions.
- In calibrating the stressed outflows from deposits, some Banks:
 - determined segmentation using a systematic bottom-up approach by analysing historical deposit data against the respective risk drivers or factors. This approach allowed the segmentation to be informed and supported by data and also facilitated better profiling of stressed outflows;
 - applied granular deposit segmentation based on liquidity risk drivers and/or factors to characterise the stability of the different segments; and
 - assumed larger outflows for segments with funding concentration risk.
- In calibrating the stressed outflows from drawdown of committed facilities, some Banks used granular segmentation based on business lines, product characteristics and/or combination of key liquidity risk drivers. For example, these Banks performed such calibration for each unique combination of business line, product, industry and credit rating.
- When calibrating the potential outflows arising from market valuation changes on derivatives, some Banks:

- used a model-based approach for forecasting collateral outflows (e.g., model additional variation margin requirements in times of stress using macroeconomic scenarios to obtain a stressed mark-to-market value);
 - catered for increased initial margin requirements and higher haircuts on collateral imposed by central clearing counterparties in times of market volatility; and
 - incorporated information from central counterparties, clearing members, such as historical margin increases during stress, when estimating increases in initial margin requirements and haircuts under extreme but plausible stress scenarios.
- In determining inflows, some Banks specified granular segmentation based on business lines and product characteristics and applied differentiated inflow treatments for these segments based on assessment of liquidity risk drivers under stress. For example, whether the business model of a segment was based on originate-to-distribute and whether a segment was cyclical to stress. For certain segments with franchise value or which performed critical economic function, some Banks assumed such loans to be continuously rolled over, resulting in higher haircut to inflows.

Assumptions on actions to generate liquidity

- Good practices observed in estimating speed and quantity of inflow assumptions from monetisation of assets under stress included:
 - using various data sources to calibrate monetisation capacity and haircuts for liquid assets, with conservatism applied where data is limited;
 - considering factors such as market depth for the asset, number of available counterparties, means of dealing (bilateral, tri-party, exchange, broker), the Bank's own credit rating and presence in the relevant markets, its historical transaction volumes, its holdings relative to normal market turnover, potential market signalling in estimating monetisation capacity;
 - applying caps on liquidity obtainable via different channels; for example, volume of total repo capacity and daily outright sales volume;
 - determining haircuts for liquid assets monetised through repo versus sales channels separately (given that the conditions in the repo and sale markets may differ); and

- widening haircuts if liquidation amount exceeds internal daily threshold.

12 Maintenance of Liquid Assets

- 12.1 A critical element of a Bank's resilience to liquidity stress is the continuous availability of an adequate buffer of unencumbered, high quality liquid assets. A Bank should maintain such assets as insurance against a range of liquidity stress scenarios, including those that involve the loss or impairment of unsecured and typically available secured funding sources. For liquid assets to be usable during crises to obtain funding, there should be no legal, regulatory or operational impediment to using these assets.
- 12.2 The size of the liquidity buffer should be commensurate with a Bank's liquidity profile and is supported by its estimates of liquidity needs under stressed conditions. It should also be aligned with the Bank's established risk appetite. A Bank should ensure that the size of its liquid asset buffer is sufficient to maintain resilience to unexpected stress, while also meeting its daily and intraday payment and settlement obligations in a timely manner throughout the duration of the stress. In doing so, the Bank should consider other available tools and resources to manage intraday liquidity risks. Please refer to section 15 for details on intraday liquidity risk management.
- 12.3 Regarding the composition of its liquidity buffer, a Bank should maintain a core of the most reliable liquid assets, such as cash and high quality government bonds or similar instruments, to mitigate the impact of severe stress scenarios. It should factor in asset liquidation uncertainty as the marketability of individual assets may differ across stress scenarios and timeframes, considering factors like the transparency of its structure and risk characteristics, ease and certainty of valuation, central bank eligibility, depth of the market for the asset, and the Bank's own name and presence in the relevant markets.
- 12.4 Considering that some stress scenarios can involve rapid outflows within a very short period, a Bank's buffer should include liquid assets that can be quickly converted into cash. To insure against less intense but longer duration stress events, a Bank may choose to broaden the composition of the buffer by including other unencumbered liquid assets that are marketable (i.e. can be sold or used as collateral in sale and repurchase agreements) without incurring excessive losses or discounts. While a Bank may include securities issued by financial institutions as part of its non-regulatory liquidity buffer, it should cater for scenarios where such securities

are likely to become illiquid, such as during a broader financial crisis. Hence, it should exercise prudence in determining the proportion of such securities in their liquidity buffers and factor this correlation risk into its assessment of the buffer's effectiveness during stress periods.

- 12.5 A Bank should carefully assess the implications of holding liquid assets classified as held-to-collect accounting classification. Selling these assets may necessitate realisation of unrealised losses, affecting the Bank's capital position. This could undermine confidence among the Bank's customers and lead to increased withdrawals, heightening liquidity risk. While these assets can be monetised through repo transactions without reporting a loss in financial statements, there is a risk that a distressed Bank may not have access to the repo market during times of stress. Therefore, it is important for the Bank to factor in this risk when determining the composition of its liquid asset portfolio. If a Bank includes held-to-collect assets in its liquid asset buffer, it should be able to demonstrate that the accounting classification does not create a barrier to its ability to monetise these assets.
- 12.6 A Bank should implement policies and guidelines to manage concentration risk within its liquid asset portfolios. These policies should address diversification across asset types, issue and issuer, and maintenance of liquid assets that are consistent with the distribution of its liquidity needs by currency.
- 12.7 A Bank should regularly assess its ability to monetise liquid assets under various channels (e.g., repo and outright sale, as well as central bank liquidity facilities²⁰) in stressed conditions. A Bank should periodically monetise a representative proportion of liquid assets (including held-to-collect assets) to test market access, to evaluate the effectiveness of the monetisation processes via different channels and to minimise the risk of negative signalling in periods of actual stress. For positions that have not been monetised in day-to-day operations, a Bank should test its market access to ascertain the liquidity of these assets and consider additional haircuts to account for uncertainties during monetisation. The results of these monetisation exercises should be used to inform the eligibility criteria for liquid assets and determine appropriate haircuts.

²⁰ For instance, MAS operates a suite of SGD liquidity facilities. Eligible counterparties for the Intraday Liquidity Facility, Standing Facility, and SGD Term Facility should ensure that they are operationally ready to utilise the facilities for liquidity management purposes. These facilities allow eligible counterparties to obtain intraday, overnight, or term SGD liquidity on a collateralised basis.

Good practices observed in monetisation of liquid assets

- Some Banks systematically monitored and reviewed the monetisation of liquid assets. They separated the portfolio of liquid assets into tested versus not tested and closely tracked the observed monetisation value and haircuts. They would also develop and implement plans to test the monetisation of previously untested assets to verify their actual liquidity values and validate their assumed market behaviour.
- Some Banks set a minimum percentage of liquid assets to be monetised annually through outright sales and/or repo to demonstrate monetisation capabilities.

12.8 During periods of significant liquidity stress, a Bank may utilise its stock of liquid assets to meet its liquidity needs, even if this results in the Bank falling below regulatory liquidity requirements.²¹ However, the Bank should subsequently develop appropriate plans to promptly rebuild its stock of liquid assets, taking into consideration the circumstances of the Bank and the prevailing economic environment.

13 Group-wide Liquidity Risk Management

13.1 A Bank should actively monitor and control liquidity risk exposures and funding needs within and across legal entities, business lines and currencies. This monitoring and control should occur at the level of individual legal entities, foreign branches and subsidiaries, as well as the group as a whole, taking into account legal, regulatory and operational limitations to the transferability of liquidity.

13.2 Each legal entity and location should have a liquidity management strategy that considers its operating environment and the extent of liquidity transferability across the group. A Bank should identify sources of funding for each location/entity/currency under both business-as-usual and contingency situations. For each jurisdiction in which it is active, a Bank should ensure that its policies and procedures account for jurisdiction-specific features of the legal and regulatory regime that influence liquidity risk management, including arrangements for dealing with failed banks, deposit insurance and central bank operational frameworks and collateral policies.

²¹ As per MAS Notice 649 for banks and MAS Notice 1015 for merchant banks.

- 13.3 To establish a comprehensive group-wide perspective of liquidity risk exposures, a Bank that has a branch or subsidiary in a foreign jurisdiction should implement processes that aggregate data across multiple systems. A Bank should monitor risk dynamics throughout the group (including at individual entity level and relevant sub-group levels where appropriate). This monitoring enables the group to have a good understanding of the impact that each legal entity or location has on the group's overall liquidity position. To analyse the impact of intragroup funding on a Bank's liquidity position, the Bank should be able to distinguish between intragroup and external cash flows.
- 13.4 A Bank should not take intragroup transferability of liquidity for granted and should implement processes to identify and monitor liquidity transfer restrictions. Assumptions on fund and collateral transferability set out in liquidity risk management policies should consider all regulatory, legal, accounting, credit, tax and internal constraints on liquidity and collateral movement. Additional constraints may emerge during stressed conditions, hence a conservative approach to setting assumptions is appropriate.
- 13.5 A Bank should monitor the legal entity and physical location of liquid assets for timely monetisation and evaluate operational arrangements needed to transfer funds and collateral across entities, including the time required to complete such transfers under those arrangements. To mitigate contagion risk and to avoid excessive reliance on funding from elsewhere in the banking group, a Bank should consider establishing internal limits or guidelines on the amount of intragroup funding allowed.
- 13.6 When a group contains both bank and non-bank entities, group level management should understand the different liquidity risk characteristics specific to each entity, with respect to both the nature of the business and the regulatory environment.

14 Individual Currency Liquidity Risk Management

- 14.1 A Bank should evaluate how currency-specific liquidity mismatches affect its overall liquidity risk exposure when determining its risk appetite for acceptable currency mismatches. Factors to consider include:
- (a) its ability to raise funds in markets for each significant currency, taking into account the depth and liquidity of these markets as well as the timing of access to such currencies and markets;

- (b) the likely extent of back-up facilities available for each significant currency in its domestic market;
- (c) its ability to transfer a liquidity surplus from one currency to another in a timely manner, and across jurisdictions and legal entities;
- (d) the likely convertibility of currencies in which it is active, including the potential for impairment or complete closure of foreign exchange swap markets for specific currency pairs; and
- (e) potential differences in behaviour between customers and counterparties across different currencies, especially under stressed conditions.

14.2 A Bank's currency mismatch risk appetite may be expressed as a limit on the size of mismatches in cash flows over specific time horizons in each significant currency under the business-as-usual scenario as well as under stress scenarios. Stricter limits should be set on currencies where the convertibility and transferability are less certain, particularly in stressed situations. Assumptions regarding the convertibility and transferability of currencies should be well supported and approved by senior management.

14.3 A Bank should undertake a separate analysis of its strategy for each significant currency, considering potential constraints in times of stress. It should implement robust risk identification, measurement, evaluation, monitoring and reporting mechanisms and controls to manage liquidity risk across individual currencies, integrated into its overall liquidity risk management framework. This includes monitoring the Bank's currency-specific risk characteristics (including understanding its funding profile, assessing deposits stability, identifying any concentration risks), managing net funding gaps against limits, stress testing, and contingency funding planning. A Bank should negotiate liquidity back-stop facilities for a specific currency and develop a broader contingency strategy, if the Bank runs significant liquidity risk positions in that currency.

14.4 A Bank should not overly rely on foreign exchange or currency swap markets for its foreign currency liquidity needs as there is a risk that its ability to swap currencies may erode rapidly under stressed conditions. This risk is especially high for currencies without fully developed foreign exchange markets. A Bank should also be mindful that changes in foreign exchange rates or market liquidity could sharply widen currency mismatches and consequent liquidity risk, as well as alter the effectiveness of foreign exchange hedges and hedging strategies. In this regard, a Bank should apply additional haircuts to liquid assets denominated in a particular currency when using these assets to fund shortfalls in another currency during stress testing. These haircuts account for potential foreign exchange depreciation and its impact on inflow values.

- 14.5 Where borrowers have currency mismatches in their own finances, this may impair their ability to service their loan obligations in the specified currency. Where this materialises, it could in turn give rise to liquidity risk for the Bank. Accordingly, where a Bank identifies that a substantial portion of its borrowers are exposed to such currency mismatch risk, it should factor this into its liquidity stress testing scenarios to better capture the potential impact on its liquidity position.

Good practices observed in individual currency liquidity risk management

Banks developed methodologies to assess their capacity to conduct foreign exchange swap for each significant currency under stressed conditions. Their assessments included the following considerations:

- Evaluating historical stress volume and net outstanding position as reference points, taking into account suitability of the historical time period chosen;
- Assessing swap capacity over various time horizons²²;
- Assessing the availability of counterparties and counterparty limits under stressed conditions, which informs total capacity²³;
- Applying additional haircuts to cater to uncertainties from cross currency funding; and
- Being aware of potential shortage of key funding currencies (e.g., USD) during market-wide stressed conditions.

15 Intraday Liquidity Risk Management

- 15.1 Intraday liquidity risk management is an integral component of a Bank's broader liquidity management strategy. A Bank should actively manage its intraday liquidity positions and risks to

²² Banks assess their foreign exchange swap capacities under stressed conditions for various time periods. This practice serves two key purposes: firstly, it helps Banks estimate the volume of swaps they can execute during different durations; secondly, it gauges their access to major currencies in both short-term and extended stress scenarios. Some Banks take a conservative approach by applying their one-day stressed foreign exchange swap capacity estimate to the entire duration of the stress test, ensuring they maintain a prudent stance throughout the assessment period.

²³ For instance, in an institutional-specific stress scenario, counterparties may be reluctant to have exposures to troubled Banks, limiting their access to foreign exchange swaps. In a market-wide stress scenario, there could be increased usage from other counterparties, constraining the Bank's market share. Both scenarios result in diminished foreign exchange swap capacity for the Bank under stress.

meet its payment and settlement obligations on a timely basis under both normal and stressed conditions.

- 15.2 Intraday liquidity risk is the risk that a Bank is unable to meet a payment obligation at the time expected. This in turn could impact its counterparties' ability to meet obligations and lead to other counterparties in the system being impacted. Delays in payments could cause other banks to postpone their own payments and thus increase uncertainty about funding needs and amplify the impact.
- 15.3 To manage intraday liquidity risk effectively, a Bank should establish a robust governance and risk management framework for intraday liquidity risk, with clearly defined roles and responsibilities for all personnel involved in managing intraday liquidity risks. Regular assessments of intraday liquidity risk drivers and monitoring of trends in intraday liquidity sources and uses (e.g., patterns, key counterparties, maximum usage levels) should be conducted. For proper management oversight, there should be discussions and reviews of intraday liquidity matters and metrics at the relevant management committee(s) regularly.
- 15.4 A Bank should develop intraday liquidity management capabilities that enable it to identify and prioritise time-specific and other critical obligations in order to meet them when expected, and settle other less critical obligations as soon as possible. The extent and sophistication of these capabilities should be commensurate with the Bank's operating model, the nature and materiality of its intraday liquidity risks, and its role in payment and settlement systems. A Bank's strategy for managing its intraday liquidity risk should minimally include the following operational elements:
 - (a) A Bank should have the capacity to measure expected daily gross liquidity inflows and outflows, anticipate the intraday timing of these flows where possible, and forecast the range of potential net funding shortfalls that might arise at different points during the day. A Bank should understand the rules of all payment and settlement systems in which they participate; identify key counterparties (and their correspondents or custodians) that act as the source of incoming or outgoing gross liquidity flows; identify key times, days and circumstances where liquidity flows and intraday credit needs might be particularly high; and understand the business needs underlying the timing of liquidity flows and intraday credit needs of internal business lines and key customers. A Bank should ask key customers, including customer banks, to forecast their own payment traffic to facilitate this process.
 - (b) A Bank should have the capacity to monitor intraday liquidity positions against expected

activities and available resources (e.g., balances held with the central bank or other banks, remaining intraday credit capacity, available collateral). Monitoring key positions frequently during the day enables a Bank to determine if actual cash flows materially deviate from initial forecasts and trigger responses such as acquiring additional intraday liquidity or limiting certain outflows where possible. EWIs should be set to detect deteriorating intraday liquidity positions, and escalation procedures should be established to facilitate timely response.

- (c) A Bank should arrange to acquire sufficient intraday funding to meet its intraday objectives. Sources of intraday funding may include intraday credit facilities provided by central banks, correspondent or custodian banks, and other market sources (e.g., by arranging for overnight money market transactions to be delivered and returned at specific times). A Bank's sources of intraday funds may need to vary within a single currency and across different currencies, especially if it has limited access to central bank intraday credit. A Bank should test the utilisation of central bank intraday facilities to ensure operational readiness if the need arises.
- (d) A Bank should have the ability to manage and mobilise collateral as necessary to obtain intraday funds (please also refer to section 17). It should have sufficient collateral available to acquire the level of intraday liquidity needed to meet its intraday objectives. It should be operationally ready to pledge or deliver this collateral to central banks, correspondents, custodians and counterparties. A Bank should also understand the timeframes required to mobilise different forms of collateral, including collateral held cross-border.
- (e) A Bank should have a robust capability to manage the timing of its liquidity outflows in line with its intraday objectives. It is important that the Bank has the ability to manage the payment outflows of key customers and, if customers are provided with intraday credit, that credit procedures are capable of supporting timely decisions. Internal coordination across business lines is important to achieving effective controls over liquidity outflows.
- (f) A Bank should be prepared to deal with unexpected disruptions to its intraday liquidity flows. As described in sections 11 and 18, a Bank's stress testing and contingency funding plans should reflect intraday considerations. It also should understand the level and timing of liquidity needs that may arise as a result of the failure-to-settle procedures of payment and settlement systems in which it is a direct participant. Robust operational risk management and business continuity arrangements are also critical to the effectiveness of a Bank's intraday liquidity management. To this end, it would be prudent for a Bank to

consider incorporating intraday liquidity stress scenarios into its contingency funding plan drill exercises. These scenarios might include issues with existing nostro agents or settlement/operational failures. Such exercises can help a Bank to better assess its ability to meet funding obligations under challenging circumstances and to test tapping intraday facilities.

- 15.5 A Bank should establish adequate policies, procedures and systems to support operational objectives in all of the financial markets and currencies in which it has significant payment and settlement flows. The tools and resources employed should be tailored to the Bank's business model, its role in the financial system, and its specific activities within each market (e.g., via direct participation in a payment or settlement system or via correspondent or custodian banks), and whether it provides correspondent or custodian services and intraday credit facilities to other banks, firms or systems. For instance, for a Bank heavily reliant on collateralised funding markets, monitoring positions in securities settlement systems is equally crucial as monitoring positions in real-time gross settlement (RTGS) systems.
- 15.6 When a Bank chooses to rely on correspondent or custodian banks for payment and settlement activities, the Bank should ensure that this arrangement allows it to meet obligations on a timely basis and to manage its intraday liquidity risks across various circumstances. It should also acknowledge the potential for operational or financial disruptions at its correspondent or custodians to impact its own liquidity management, and have alternative arrangements in place to ensure it can continue to meet its obligations in such situations.
- 15.7 For monitoring of intraday liquidity profile, Banks should refer to the BCBS' guidance on the intraday liquidity monitoring tools in SRP50 Liquidity Monitoring Metrics²⁴ in the Basel Framework as a reference for developing appropriate monitoring capabilities. A Bank should implement monitoring tools that are relevant to its intraday liquidity risks. The SRP50 metrics include the following:

Tools applicable to all Banks

- (a) Daily maximum intraday liquidity usage;
- (b) Available intraday liquidity at the start of the business day;
- (c) Total payments;
- (d) Time specific obligations;

²⁴ https://www.bis.org/basel_framework/chapter/SRP/50.htm

Tools applicable to Banks that provide correspondent banking services

- (e) Value of payments made on behalf of correspondent banking customers;
- (f) Intraday credit lines extended to customers;

Tools applicable to direct participants of RTGS systems

- (g) Intraday throughput.

- 15.8 Please refer to SRP50 Liquidity Monitoring Metrics in the Basel Framework for detailed explanations of each monitoring tool. Each tool has individual informational value, and tracking multiple tools will provide a more comprehensive view of a Bank's resilience to intraday liquidity shocks.
- 15.9 Effective intraday liquidity risk management requires seamless coordination across a Bank's front, middle and back offices (e.g., treasury, business units, and operations/settlement teams). This involves close monitoring of anticipated payments, swift investigation of payment delays and management of unexpected intraday cash flows. Given the time-critical nature of meeting settlement deadlines, clear delineation of tasks and responsibilities is crucial to ensure all personnel understand their roles. Robust escalation protocols should be in place to address settlement issues promptly, while all personnel should be prepared to take prompt action when required.

Stress Test

- 15.10 As the availability and usage of intraday liquidity can change significantly in times of stress, it is important for a Bank to consider the impact of stressed conditions on its intraday liquidity requirements. A Bank should conduct regular stress testing on a variety of stress scenarios (drawing on historical experience/stress events and forward-looking analyses) that can affect intraday liquidity positions as well as the availability of intraday liquidity resources. It should use intraday stress test results to inform management decisions (e.g., intraday liquidity risk tolerance and buffer levels, adequacy of intraday liquidity processes, including potentially the development of back-up service arrangements to tackle potential disruptions, and contingency funding plan).
- 15.11 As a guidance, four possible (non-exhaustive) stress scenarios have been identified as follows:

- (a) Own financial stress: a Bank suffers or is perceived to be suffering from a stress event. For a direct participant, own financial and/or operational stress may result in counterparties deferring payments and/or withdrawing intraday credit lines. This, in turn, may result in the Bank having to fund more of its payments from its own intraday liquidity sources to avoid having to defer its own payments. For a Bank that uses correspondent banking services, an own financial stress may result in intraday credit lines being withdrawn by the correspondent bank(s), and/or its own counterparties deferring payments. This may require the Bank having either to prefund its payments and/or to collateralise its intraday credit lines. In the context of digital banking, potential large-scale, rapid withdrawals could quickly deplete a Bank's intraday liquidity, potentially leading to cascading effects on its ability to meet other payment obligations.
- (b) Counterparty stress: a major counterparty or major counterparties suffers/suffer an intraday stress event which prevents it/them from making payments, resulting in a Bank being unable to rely on incoming payments from the stressed counterparty/counterparties.
- (c) A customer bank's stress: a customer bank of a correspondent bank suffers a stress event. As a result, there could be deferment of payments to the customer, creating a further loss of intraday liquidity at its correspondent bank.
- (d) Market-wide credit or liquidity stress: this may have adverse implications for the value of liquid assets that a Bank holds to meet its intraday liquidity usage. A widespread fall in the market value and/or credit rating of a Bank's unencumbered liquid assets may constrain its ability to raise intraday liquidity from the central bank or correspondent banks. A Bank which manages intraday liquidity on a cross-currency basis should consider the intraday liquidity implications of a closure of, or operational difficulties in, currency swap markets and stresses occurring in multiple systems simultaneously.

- 15.12 For the own financial stress and counterparty stress scenarios, a Bank should consider the likely impact that these stress scenarios would have on its daily maximum intraday liquidity usage, available intraday liquidity at the start of the business day, total payments and time-specific obligations.
- 15.13 For the customer bank's stress scenario, a Bank that provides correspondent banking services should consider the likely impact that this stress scenario would have on the value of payments made on behalf of its customers and intraday credit lines extended to its customers.

- 15.14 For the market-wide stress, a Bank should consider the likely impact that the stress would have on its sources of available intraday liquidity at the start of the business day.

- 15.15 Where relevant, a Bank should also be aware that stress events may simultaneously give rise to time-critical liquidity needs in multiple currencies and multiple payment and settlement systems. Those liquidity needs could arise from both the Bank's own activities as well as those of its customer banks and firms. They also could arise from the special roles a Bank might play in a given settlement system, such as acting as a back-up liquidity provider or settlement bank.

- 15.16 A Bank should be prepared for potential increases in intraday liquidity requirements in periods of stress. These increases may arise from changes in counterparty payment behaviour, such as delayed payments, and/or increased pre-positioning requirements imposed on the Bank under stress.

- 15.17 While pledges of assets for these purposes typically involve intraday encumbrances that are released at the end of a settlement cycle, persistent intraday needs can impact the availability of assets to meet ongoing liquidity needs. To mitigate this risk, it is prudent for Banks to maintain additional liquid assets as a buffer for intraday liquidity risks and consider intraday liquidity usage in their ongoing stress tests.

- 15.18 For a foreign bank branch with centralised liquidity management, it should assess whether intraday liquidity risk is material to its Singapore operations. If the assessment demonstrates very limited intraday liquidity risk (e.g., due to minimal time-specific obligations, pre-funding of payments), a less extensive intraday liquidity risk management framework may be needed. Group-level consolidated stress testing may be suitable if it adequately captures and addresses the branch's intraday liquidity risks.

Buffer for Meeting Intraday liquidity

- 15.19 A Bank should maintain sufficient liquidity buffer for meeting intraday liquidity needs under both business-as-usual and stressed conditions. This buffer should comprise, at a minimum, readily available cash or pre-pledged assets at central banks (where relevant) and relevant institutions (such as correspondent banks, or payment or settlement systems). The size of the buffer should be calibrated to account for factors such as uncertainties in intraday liquidity flows, holidays, potential unexpected events, and anticipated cash flow behaviour under stressed conditions. It should also be informed by the Bank's stress test.

16 Diversification and Management of Market Access

- 16.1 A Bank should establish funding strategies that provide effective diversification in the sources (e.g., counterparties, instruments, and markets) and tenor of funding, spanning short-, medium- and long-term horizons. Funding plans should take into account the possibility that different sources of funds may be exposed to the same underlying market conditions, causing them to become unavailable at the same time. For instance, funding from counterparties that share common characteristics, such as operating in the same industry or relying on the same underlying investor base, may dry up simultaneously when market conditions change and adversely affect that segment as a whole. Therefore, a Bank should consider diversifying funding sources across dimensions that are less likely to be correlated.
- 16.2 Funding concentration risk arises when a Bank is heavily dependent on specific sources for its funding. The disruption of such concentrated funding sources could trigger significant withdrawals, especially when funding providers share common characteristics or exhibit herding behaviour, thereby threatening a Bank's liquidity position. Concentration risks can arise from various dimensions, including the type of funding providers (e.g., industry, geography), tenor, instrument (e.g., bonds, commercial paper), market, currencies and others. Excessive concentration could lead to substantial withdrawals due to the impact of a single factor or herding behaviour.
- 16.3 To mitigate concentration risk, a Bank should monitor funding concentration across various dimensions. A Bank should establish limits where appropriate to control concentration risks and review these limits regularly to ensure they remain relevant to changing market conditions and the Bank's risk appetite. A Bank that relies on wholesale funding, which tends to be more volatile than retail funding²⁵, should ensure that its funding sources are sufficiently diversified to maintain timely availability of funds at the right maturities and at reasonable costs. A Bank should also monitor the concentration of cash inflows across wholesale customers as part of its liquidity management in order to limit overreliance on the arrival of cash inflows from one or a limited number of wholesale customers on a given day.
- 16.4 To ensure adequate market access and effective funding diversification, a Bank should establish

²⁵ While retail funding is generally considered more stable than wholesale funding, not all retail funding segments exhibit the same degree of stability. Certain retail depositor segments, such as high-net-worth individuals, can exhibit higher flightiness, and concentration within such segments may expose a Bank to heightened liquidity risk. A Bank should therefore account for these differences when assessing its funding profile.

and maintain an active presence in its chosen funding markets and strong relationships with funds providers. Regular market participation during normal conditions is crucial, to ensure familiarity and operational readiness, as well as to avoid negative market perceptions that may come with accessing markets only during stress periods. To establish and sustain such market participation, the Bank should develop and maintain:

- (a) strong relationships with current and potential fund providers, including central banks where appropriate;
- (b) practical dealing experience and market expertise;
- (c) adequate operational and legal arrangements, including up-to-date legal documentation; and
- (d) established systems and procedures for accessing these markets.

16.5 A Bank should regularly assess its capacity to raise funds quickly from each source by identifying and monitoring key factors that could affect its ability to do so. This assessment should be supported by a comprehensive understanding of the legal framework governing its various funding channels and markets.

16.6 Notwithstanding these established funding arrangements, in times of stress, even normally reliable funding markets can be disrupted. Hence, a Bank should take a prudent view of its relationship with fund providers and identify alternative sources of funding. These alternative sources may include deposit growth, lengthening of maturities of liabilities, issuing new short- and long-term debt instruments, intragroup fund transfers, new capital issues, sale of subsidiaries or lines of business, asset securitisation, sale or repo of unencumbered, highly liquid assets, drawing down committed facilities, and borrowing from the central banks' lending facilities. A Bank should regularly review and test these fund-raising options to evaluate its effectiveness at providing liquidity in the short-, medium- and long-term horizons.

17 Asset Encumbrance and Collateral Management

Asset Encumbrance

17.1 A Bank should maintain a comprehensive overview of its asset encumbrance level. Encumbered assets are assets that a Bank is restricted or prevented from liquidating, selling, transferring or assigning due to legal, regulatory, contractual or other limitations.

- 17.2 A high level of asset encumbrance means that a Bank has less capacity to raise additional funding during stress periods with the remaining assets. This reduced flexibility can lead to higher funding costs as creditors demand larger spreads and may react more sensitively to stress signals. Hence, it can impact a Bank's funding stability.
- 17.3 To mitigate associated risks, a Bank should manage encumbered balance sheet assets within appropriate limits, considering both the impact on funding costs and long-term liquidity sustainability.

Collateral Management

- 17.4 A Bank should have the ability to calculate the value of all its collateral positions, including assets currently pledged relative to the amount of security required and unencumbered assets available to be pledged as collateral.
- 17.5 A Bank should monitor its level of available collateral by legal entity, jurisdiction and currency exposure and be aware of the operational and timing requirements for accessing collateral given its physical location (including the custodian bank or securities settlement system with which the collateral is held).
- 17.6 A Bank should assess the eligibility of each major asset class for pledging as collateral with central banks (for intraday credit, overnight and term lending operations, and borrowing under standing facilities) and the acceptability of assets to major counterparties and fund providers in secured funding markets.
- 17.7 A Bank should diversify its sources of collateral, taking into consideration capacity constraints, name-specific concentrations, the sensitivity of prices, haircuts and collateral requirements under conditions of name-specific and market-wide stress, and the availability of funds from private sector counterparties in various market stress scenarios. A Bank should be particularly vigilant about the risk that its pledged assets may decline in market value at the same time the Bank experiences liquidity stress if these two events are highly correlated.
- 17.8 A Bank should adjust, as necessary, measures of available collateral to account for assets that are part of a "tied position" (e.g., assets used as part of a hedge of an off-balance sheet or derivative position, such as an equity/debt position as a hedge to a total return swap or a negative basis trade). It should have a detailed understanding of, and be able to demonstrate, the estimated period of time to liquidate those assets or put on a substitute hedge.

- 17.9 A Bank should have sufficient collateral to meet expected and unexpected borrowing needs and potential increases in margin requirements over different timeframes (including intraday, short term and longer term structural liquidity requirements), depending upon the Bank's funding profile. Its system should be capable of monitoring shifts between intraday and overnight or term collateral usage. This is important as a given asset can only provide collateral support for only one type of credit facility at a time, creating the need for effective collateral management given competing demands.
- 17.10 In determining the level of collateral to pledge or deliver for intraday liquidity needs, a Bank should consider the potential for significant uncertainty around the timing of intraday flows within the day as well as potential operational and liquidity disruptions that could necessitate the pledging or delivery of additional intraday collateral.
- 17.11 A Bank using derivatives should consider the potential for contractually specified additional collateral requirements due to changes in market valuations, changes in the Bank's credit rating, or financial condition. It should establish appropriate collateral arrangements to meet margin and collateral calls and monitor other trigger events that could require hypothecation or delivery of additional assets to the pool. For example, a Bank that receives funding through the securitisation of a pool of assets (e.g., residential mortgages or credit card receivables) should monitor embedded trigger events that could necessitate hypothecating or delivering additional assets to the pool. A Bank's MIS should be able to report whether it has sufficient unencumbered assets of the right type and quality for such contingencies.
- 17.12 A Bank active in secured funding and derivatives should maintain robust operational processes and collateral management practices²⁶, such as the following:
- (a) regularly reviewing cash and collateral management systems to ensure they are well-designed, operationally resilient and fit for purpose;
 - (b) conducting exercises to test operational capabilities to meet cash and collateral calls under stress, including on an intraday basis;
 - (c) maintaining active, transparent and regular interactions with their counterparties and third party service providers of collateral management services;
 - (d) considering the advantages of standardising and automating collateral management processes to reduce frictions and the possibility of operational delays or failures in collateral

²⁶ For additional guidance on this topic, a Bank may refer to the Liquidity Preparedness for Margin and Collateral Calls: Final Report issued by the Financial Stability Board on 10 December 2024 (<https://www.fsb.org/2024/12/liquidity-preparedness-for-margin-and-collateral-calls-final-report/>).

- use, especially during stress periods;
- (e) implementing clear dispute mechanisms and a system to track and manage outstanding collateral disputes; and
- (f) evaluating the appropriate levels and quality of excess collateral posted in non-stress times, when considering the pro-cyclical nature of margining and collateralisation mechanics.

18 Contingency Funding Plan

- 18.1 A contingency funding plan (CFP) is the compilation of policies, procedures and action plans for responding to severe disruptions to a Bank's ability to fund some or all of its activities in a timely manner and at a reasonable cost. A Bank should establish a robust CFP that clearly sets out strategies for addressing liquidity shortfalls in emergency situations, without relying on central bank emergency liquidity assistance²⁷.
- 18.2 When developing its CFP, a Bank should ensure that the plan is commensurate with the nature, size and complexity of its activities and its role in the financial system and takes into account the Bank's unique characteristics. A Bank with a simpler business model may develop a more streamlined CFP whilst still addressing all key components outlined in this section.
- 18.3 The CFP should be formally documented. A Bank should include in its CFP a clear description of diverse, viable and readily available contingency funding measures for preserving liquidity and addressing cash flow shortfalls in a range of stress environments. The CFP should provide a framework with a high degree of flexibility so that a Bank can respond quickly in a variety of situations.
- 18.4 The CFP's design and procedures should be closely integrated with a Bank's ongoing liquidity risk analysis, and with the results of scenarios and assumptions used in stress tests, addressing issues over a range of time horizons, including intraday.
- 18.5 An effective CFP should comprise the following components:
- (a) roles and responsibilities, decision-making and escalation procedures during a liquidity crisis;

²⁷ Emergency liquidity assistance (ELA) generally refers to obtaining liquidity from MAS or other central banks (if applicable) that goes beyond the standing liquidity facilities operated by them. For information on MAS' ELA framework, a Bank may refer to MAS' ELA Monograph (<https://www.mas.gov.sg/-/media/mas/monetary-policy-and-economics/central-bank-operations-and-liquidity-management/emergency-liquidity-assistance-monograph.pdf>).

- (b) funding options and action plans;
- (c) liquidity reporting requirements during a crisis;
- (d) communication plans; and
- (e) testing, updating and maintaining CFP.

Roles and Responsibilities, Decision-Making and Escalation Procedures During a Liquidity Crisis

- 18.6 A Bank's CFP should have clear policies and procedures that enable timely, well-informed decision-making, and swift execution of contingency measures. The CFP should provide clarity on roles and responsibilities, including the authority to invoke the CFP and the composition of the formal "crisis team" to facilitate internal coordination.²⁸ The CFP should also set out a clear decision-making process and escalation procedures to be implemented during periods of liquidity stress.²⁹
- 18.7 There should be clear procedures for monitoring, escalation, and activation. When EWIs signal potential stress, senior management should promptly assess the situation and determine whether to activate the CFP. The criteria for CFP activation should be pre-determined and will likely be based on a combination of quantitative triggers and qualitative judgment. It is important not to be overly-reliant on expert judgement as that could lead to delays in CFP activation.
- 18.8 A Bank should have its EWI monitoring and escalation procedures independently reviewed and tested. Such reviews and testing may be conducted by independent parties such as risk management functions (if they are not involved in the design of the EWIs), internal or external auditors, or other third parties with relevant expertise. These reviews should assess the adequacy of the adopted EWIs, the appropriateness of the trigger thresholds, and the effectiveness of monitoring and escalation procedures (e.g., when a threshold is triggered). A Bank should regularly review and update its EWIs and the triggering thresholds where appropriate.

Funding Options and Action Plans

- 18.9 A CFP should prepare a Bank to manage a range of severe liquidity stress scenarios. These should

²⁸ There should be a list of names, contact details and locations of team members responsible for implementing the CFP, as well as designated alternates for key roles.

²⁹ These include a decision-making process on what actions to take at what time, who can take them and what issues need to be escalated to more senior levels in the Bank, including procedures for coordination across business lines and locations.

encompass both firm-specific and more generalised market-wide stress, as well as the potential interaction between them over various time horizons. For this purpose, a Bank's CFP should contain a diverse menu of options, such as sources of additional funding for each significant currency and plans to alter balance sheet asset and liability characteristics.

- 18.10 There should be a prioritisation framework for determining which options are suited to given circumstances, considering impediments and dependencies. For each funding source or option, there should be information pertaining to:
- (a) detailed execution procedures³⁰;
 - (b) circumstances under which each option can be taken;
 - (c) lead time needed to tap funds from each source;
 - (d) amount of funds estimated that can be tapped from each option;
 - (e) profit/loss and capital impact, including any possibility of crystallising unrealised profit/loss (e.g., from liquidation of a Bank's investment portfolios that are accounted for on a held-to-collect basis);
 - (f) any operational hurdles;
 - (g) any reputational impacts from tapping each source; and
 - (h) potential measures that can be taken to address operational hurdles or reputational impact.
- 18.11 A Bank should take a conservative approach to assuming the values, timing and financial impacts of contingency funding options, recognising that these parameters are highly uncertain during crises. It should support these estimates with proper justification and should critically assess the feasibility of identified potential funding sources during stress periods. When designing its CFP, it should account for:
- (a) the impact of stressed market conditions on its ability to sell or securitise assets;
 - (b) the link between asset market and funding liquidity (e.g., extensive or complete loss of typically available market funding options);
 - (c) potential changes in counterparty's risk appetites towards the Bank (e.g., the reliability and accessibility of credit lines and committed facilities during crises, whether relationships with fund providers can be maintained during stress, and the impact of reputational issues on funding access);
 - (d) subsequent rounds and reputational effects related to execution of contingency funding measures; and

³⁰ Such procedures should incorporate lessons learnt from testing and/or actual execution of contingency funding options, such as potential market dynamics and operational interdependencies.

- (e) the potential to transfer liquidity across group entities, borders and lines of business, considering legal, regulatory, operational, time zone and internal restrictions and constraints.

- 18.12 A Bank should establish systems and processes for monitoring and executing contingency funding options to determine whether the option is having its desired effect and whether there are unintended consequences. This will facilitate adjustments to the plans, if necessary.
- 18.13 A Bank should ensure that relevant staff are aware of the operational procedures needed to transfer liquidity and collateral³¹ across different locations, entities and systems, and the restrictions governing such transfers. The Bank should incorporate realistic timelines for such transfers into its liquidity estimations. It should ensure that assets intended for pledging as collateral when utilising back-up funding sources should be in a legal entity and location consistent with the Bank's funding plans.

Central bank lending facilities

- 18.14 A Bank should be conversant in all relevant central bank lending programmes, including types of lending facilities, acceptable collateral, operational procedures for accessing central bank funds and potential reputational issues associated with their use, if any. To maintain operational readiness, a Bank should regularly monitor and update collateral eligibility, collateral capacity and required legal documentation in its CFP, and test their access to these facilities to ensure that the necessary systems, processes and documentation are in place. These preparatory measures enable a Bank to swiftly access central bank liquidity facilities when needed.

Intraday liquidity needs

- 18.15 When intraday liquidity resources become scarce, a Bank should be able to identify critical payments, schedule payments based on priority, and acquire additional sources of intraday liquidity, including by identifying and mobilising additional collateral. A Bank's CFP should recognise that time-critical settlement needs may arise not only from the Bank's own transactions, but also from its customers, and from its provision of services to payment and settlement systems (e.g., by acting as a contingency liquidity provider). In a name-specific crisis, it may face increased liquidity requirements (e.g., from payment agents, central counterparties and clearing banks to preposition collateral to facilitate payment settlements). A Bank should

³¹ For instance, how different types of collateral are handled, and whether there are contractual agreements in place.

develop a CFP that is sufficiently robust to handle simultaneous disruptions in multiple payment and settlement systems.

Channel network management and bank run procedures

- 18.16 A Bank should incorporate comprehensive measures in its CFP to address potential deposit runs through both physical and digital channels, whether occurring independently or simultaneously. The bank run management procedures should include robust mechanisms to detect and manage sudden outflows across all channels with clear delineation of roles and responsibilities for staff during and outside business hours to reflect the 24/7 nature of current banking operations. These procedures should encompass operational readiness measures such as liquidity mobilisation, contingency arrangements at physical locations to handle increased customer volume (e.g., queue management, note replenishment where applicable) and to maintain heightened monitoring of IT system capacity, and clear protocols for handling stakeholder concerns.

Cross jurisdiction and foreign bank operations

- 18.17 A Bank that operates in multiple jurisdictions should have a coherent group-wide CFP that is able to meet the needs of its various entities. The Bank should assess and account for interdependencies between location-specific CFPs in its planning.
- 18.18 A foreign bank's branch or subsidiary in Singapore should maintain a CFP for its Singapore operations. The CFP for a foreign bank branch may reference or build on the head office CFP, but should include details that are Singapore-specific. For example, the CFP should describe how the local operation works with the head office/parent/group in liquidity crisis management. This description should include the extent to which the Singapore operation's liquidity is supported by liquid assets held or managed locally, and the degree of commitment from the head office/parent/group to provide liquidity support under stress situations, especially when other parts of the group also require funding at the same time.

Liquidity Reporting Requirements During a Crisis

- 18.19 A Bank should have the capability to assess its liquidity position on an ongoing basis. To this end, the Bank should incorporate into its CFP specific procedures and reporting requirements to ensure timely information flows to senior management to support its decision-making process in a crisis. The specific reports required should be tailored to the Bank's business model,

complexity and risk profile. Some examples of liquidity management reports are as follows:

- (a) cash flow reports by currency;
- (b) liquid asset holdings by currency, product type, location held and legal entity (if relevant)
- (c) cash positions;
- (d) deposit movement analysis at overall and granular levels, (e.g., covering branches, ATM locations, digital channels, currencies and depositor categories);
- (e) maturity schedule for wholesale funding and fixed deposits in the near term;
- (f) top depositors/fund providers by currency and counterparties;
- (g) projection of loan disbursement and repayments for the upcoming month;
- (h) undrawn commitment levels by business lines and currencies;
- (i) funding capacity reports by source type;
- (j) collateral availability reports that enable tracking and monitoring of eligible collateral to secure backup liquidity facilities (from both private sources and central banks);
- (k) margin call and payment reports (i.e. reports that track the margin calls that need to be made or received, including both initial and variation margin);
- (l) intraday liquidity reports by currency (e.g., time-sensitive payments that will take place within the day);
- (m) intragroup lending and borrowing report;
- (n) internal stress test reports; and
- (o) LCR projections.

Communication Plans

- 18.20 A Bank should develop communication plans for various stress scenarios as part of its CFP. These plans should deliver timely, clear, consistent and frequent communication to internal parties (such as employees across different business lines and locations) as well as external parties (including market participants, clients, creditors, shareholders, system operators, MAS and other relevant local or overseas authorities). There should be clear and consistent communication strategies across the group. The Bank's senior management should review such plans at least once a year.
- 18.21 A Bank should include in its communication plans:
- (a) specified roles and responsibilities of management and a dedicated communications team responsible for managing communication with various stakeholders during crises;
 - (b) clearly identified stakeholders to communicate with, including specific names and contact information (rather than generic titles) for all significant internal and external parties;

- (c) procedures to deliver communications within the Bank (including across different business lines and locations) and with external stakeholders (e.g., addressing when and how to contact external parties, including banking supervisors, central banks and payment system operators, correspondent banks, custodians, counterparties and customers), recognising their actions could significantly affect the Bank's liquidity position and may vary with the underlying source of a problem; and
- (d) a comprehensive crisis communication plan with pre-drafted holding statements and reply templates to staff, customers, banking supervisors and media (including strategies to address social media rumours and customer panic), to maintain confidence during crisis situations.

Testing, Updating and Maintaining CFP

18.22 A Bank should be operationally ready to execute its CFP at all times. To achieve this, the Bank should have a comprehensive testing framework to regularly review and test its CFP. This framework should have clear objectives, testing formats and coverage parameters (e.g., specific business/operation units, local entity level or group/overseas entities level). The testing plan should ensure that all key aspects are examined over a reasonable period of time. The testing exercise and its results should be documented. For contingent funding options, the Bank should document the required testing frequency and methodology for all contingent funding options and conduct these tests accordingly.

18.23 Key aspects of testing may include:

- (a) Assessing a Bank's responses to various stress scenarios (including idiosyncratic, market-wide or combined)

These scenarios should be tailored to the Bank's business model, location, and client base. They are designed to trigger management discussions on potential actions, identify possible impacts and highlight information gaps. The scenarios should be sufficiently severe to allow management to practise crisis response in a fast-changing operating environment. This approach helps to draw out valuable lessons and identify potential challenges and implications in taking certain crisis responses.

- (b) Testing how well a Bank's staff understand their roles and responsibilities

A Bank should verify that its staff's roles and responsibilities are appropriate and understood, and that contact information is up-to-date. The Bank should test contact procedures for key members or their deputies. This includes assessing interactions with key personnel at the

overseas entities or head office/parent/group for necessary coordination, approval, or support under different stress scenarios.

(c) Testing end-to-end processes which may be implemented during CFP activation to verify a Bank's operational readiness and staff awareness

These processes may include:

- generating key liquidity reports and accessing relevant systems for monitoring liquidity positions to facilitate prompt information availability;
- executing contingency funding options related to private markets (e.g., selling or repoing assets, entering into foreign exchange swaps or drawing down credit lines);
- accessing central bank liquidity facilities, including posting/pledging of a range of acceptable collaterals;
- topping up of Net Debit Cap for Fast And Secure Transfers (FAST) transfers³²;
- reviewing necessary legal and operational documentation for swift execution;
- transferring cash and collateral (especially across jurisdictions and entities) and checking ability to track its location and report the amount available in near real time; and
- moving and posting collateral under simulated adverse conditions (e.g., close to the end of the day), to verify operational readiness.

(d) Evaluating and testing of communication plans

These processes may include:

- evaluating communication scripts, drawer statements, frequently-asked-questions or press release/conferences materials prepared for internal and external parties to ensure applicability to stress scenarios; and
- testing information dissemination via different channels.

18.24 Senior management should review all aspects of the CFP following each exercise and ensure that follow-up actions are completed. Lessons learnt from such testing exercises should be shared amongst entities, such as from the head office/parent to overseas entities or from a local entity to the head office/parent and the rest of the group, where applicable. Reviews and updates to the CFP should be submitted for the board's approval at least annually, or more often, as

³² Fast And Secure Transfers (FAST) is an electronic funds transfer service that enables instantaneous transactions between major participating banks and non-bank financial institutions in Singapore. The Net Debit Cap is the maximum debit position that each FAST participant sets in the FAST system and this corresponds with the Cash Custody Account with MAS. Once a FAST participant's net debit position reaches its Net Debit Cap, further outgoing FAST transactions would be rejected by the FAST system until its Net Debit Cap exceeds the net debit position again.



changing business or market circumstances require.

- 18.25 A Bank should align its CFP with its business continuity plans (BCP) to ensure that it is still able to execute its CFP if its BCP has been triggered as well. As such, the Bank should ensure effective coordination between teams managing issues surrounding liquidity crises and business continuity. The Bank should provide liquidity crisis team members and alternates with ready access to CFPs on- and off-site. It should maintain CFPs in a corporate central repository as well as at locations that would facilitate quick implementation by responsible parties under crisis situations.